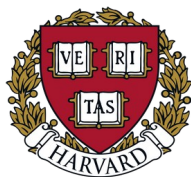




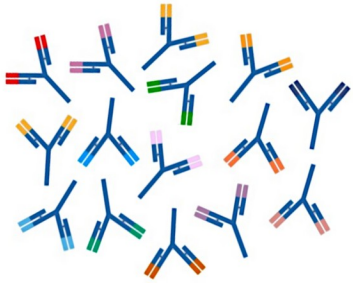
Dynamics of the spike-specific B-cell repertoire following SARS-CoV-2 Vaccination

Thomas Duplic

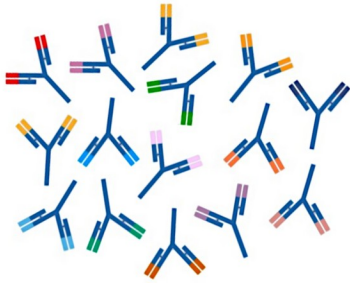
April 10th 2026, MiLaboratories



The B-cell Repertoire

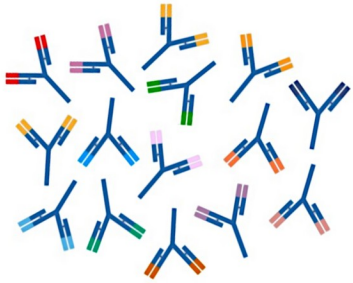


The B-cell Repertoire



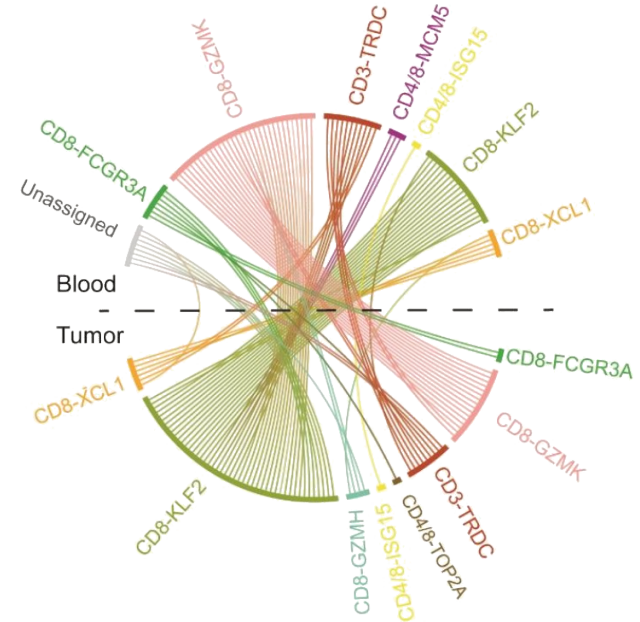
Extremely high diversity, at least 10^9 different receptors in one individual.

The B-cell Repertoire

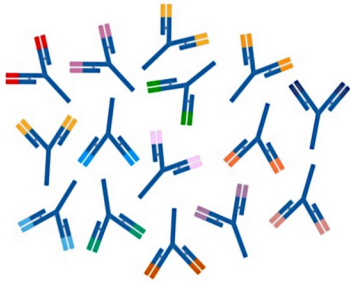


Extremely high diversity, at least 10^9 different receptors in one individual.

BCR genes are easily sequenced from bulk and single-cell RNA (RepSeq).

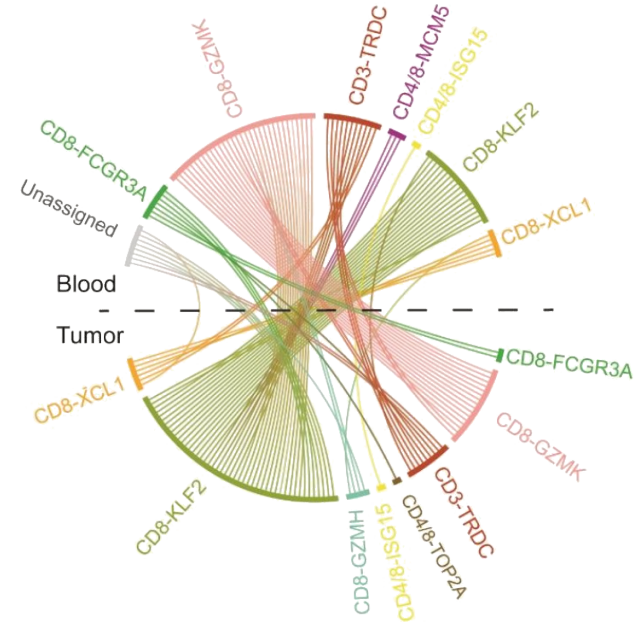
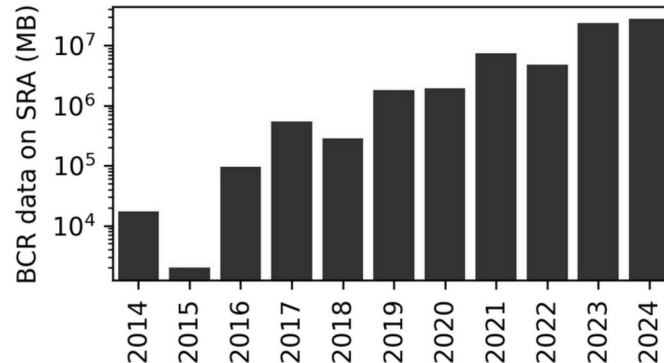


The B-cell Repertoire



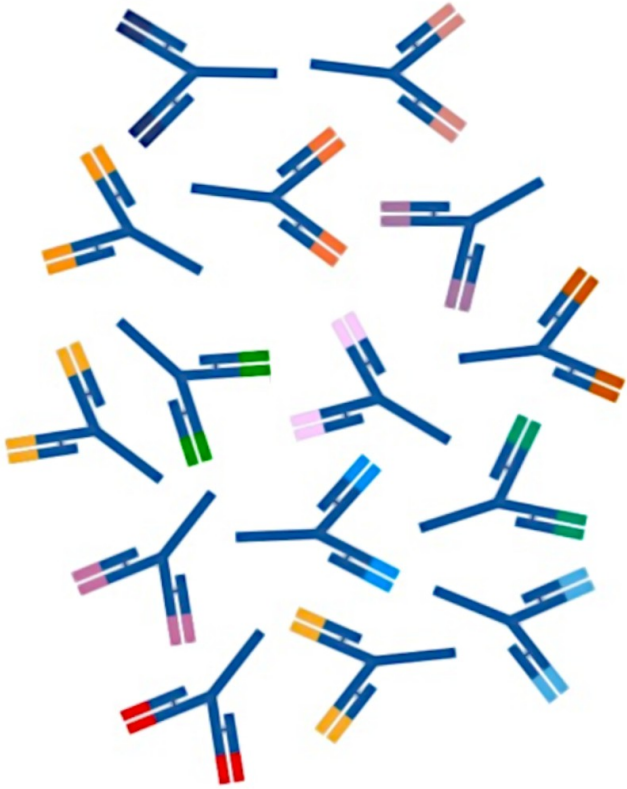
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The Potential of the Repertoire

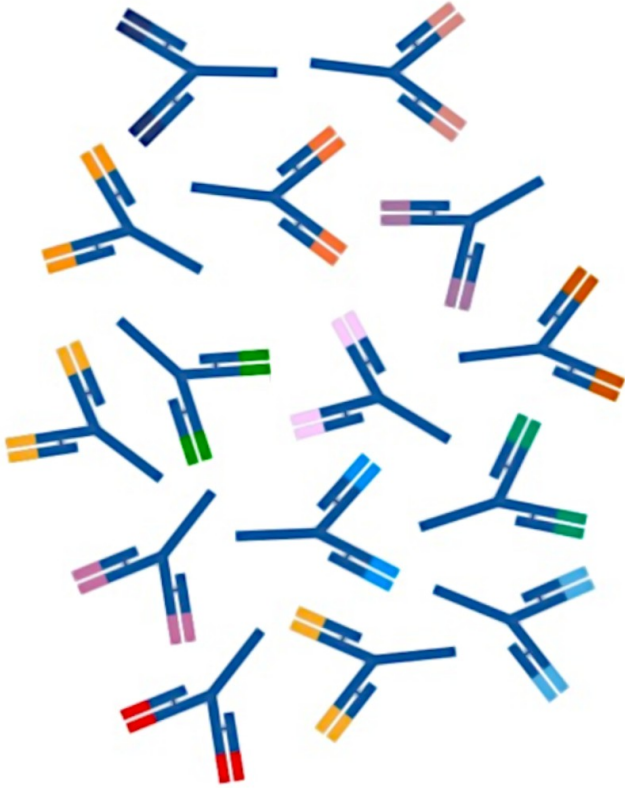
⇒ Prognosis



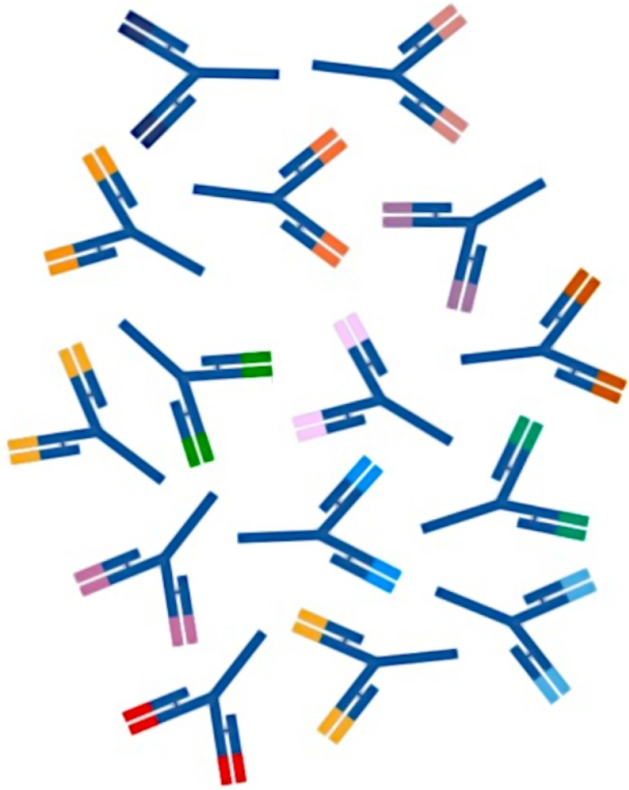
The Potential of the Repertoire

⇒ Prognosis

⇒ Diagnostic & Medical history



The Potential of the Repertoire

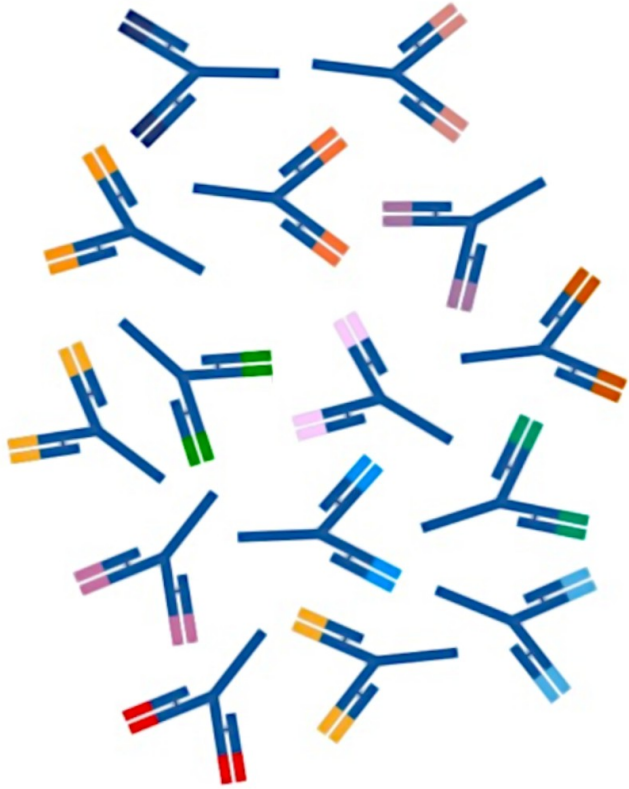


⇒ Prognosis

⇒ Diagnostic & Medical history

⇒ Therapeutic Antibodies

The Potential of the Repertoire



⇒ Prognosis

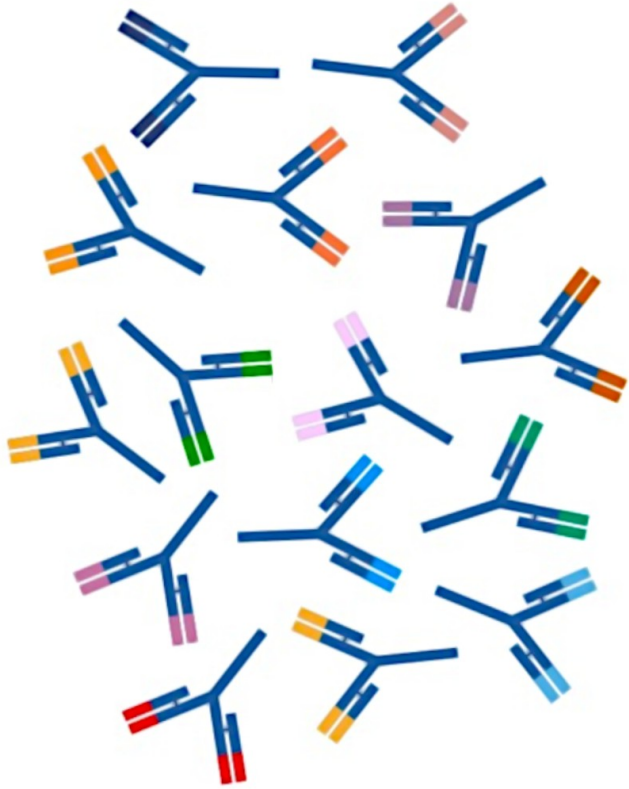
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Neoantigen quality predicts immunoediting in survivors of pancreatic cancer

[Marta Łuksza](#) , [Zachary M. Sethna](#), [Luis A. Rojas](#), [Jayon Lihm](#), [Barbara Bravi](#), [Yuval Elhanati](#), [Kevin Soares](#), [Masataka Amisaki](#), [Anton Dobrin](#), [David Hoyos](#), [Pablo Guasp](#), [Abderezak Zebboudj](#), [Rebecca Yu](#), [Adrienne Kaya Chandra](#), [Theresa Waters](#), [Zagaa Odgerel](#), [Joanne Leung](#), [Rajya Kappagantula](#), [Alvin Makohon-Moore](#), [Amber Johns](#), [Anthony Gill](#), [Mathieu Gigoux](#), [Jedd Wolchok](#), [Taha Merghoub](#), ... [Vinod P. Balachandran](#)

⇒ Diagnostic & Medical history

⇒ Therapeutic Antibodies



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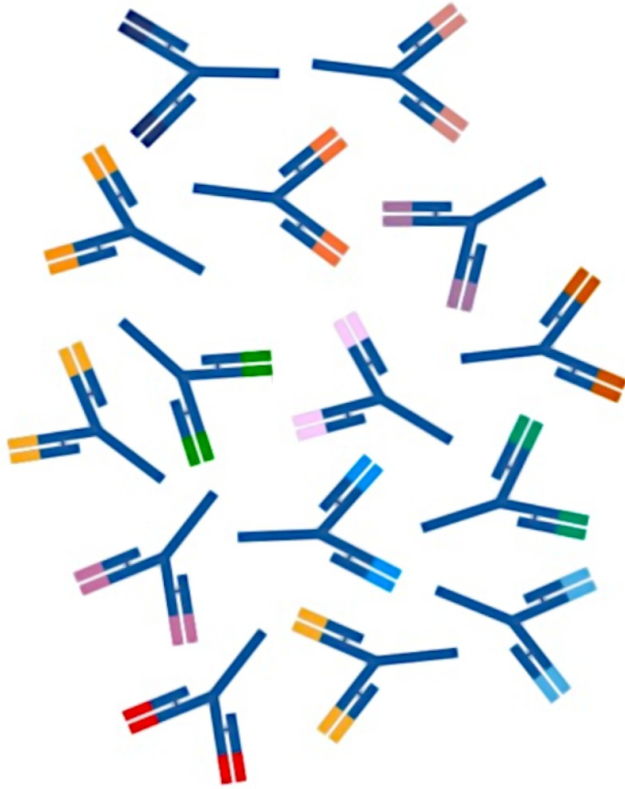
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Disease diagnostics using machine learning of B cell and T cell receptor sequences

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⇒ Therapeutic Antibodies

Functional HIV-1/HCV cross-reactive antibodies isolated from a chronically co-infected donor

[Kelsey A. Pilewski](#) ^{1,2,17} · [Steven Wall](#) ^{1,2} · [Simone I. Richardson](#) ^{3,4} · ... · [Spyros Kalams](#) ² · [Lynn Morris](#) ^{3,4} · [Ivelin S. Georgiev](#)  ^{1,2,13,14,15,16,19}  ... [Show more](#)

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⇒ Diagnostic & Medical history
Why are we not there yet ?

Disease diagnostics using machine learning of B cell and T cell receptor sequences

[MARTIN E. PAULSEN](#) , [JUN LI](#) , [ANDREW A. MATHIS](#) , [JENNIFER L. HANSEN](#) , [JUN-YEUN LEE](#) , [KHOA D. NGUYEN](#) , [RAMONA A. HOH](#) 
+27 authors · [Authors info & affiliations](#)

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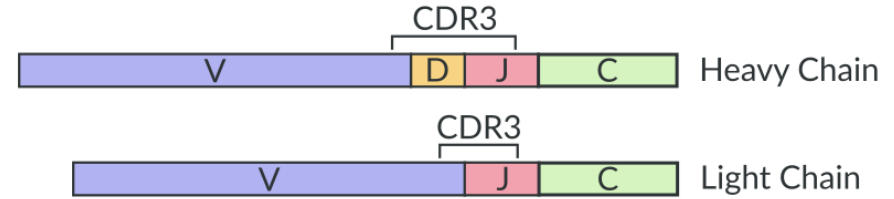
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Statistical Problems

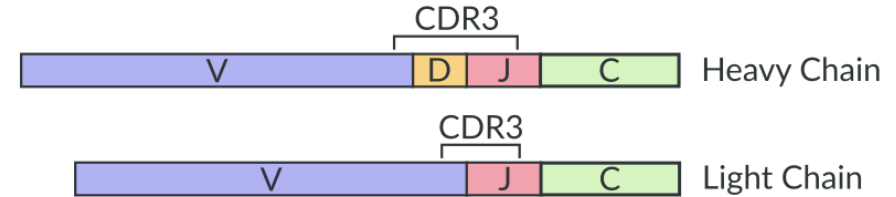
Statistical Problems

BCRs are randomly generated by V(D)J recombination.

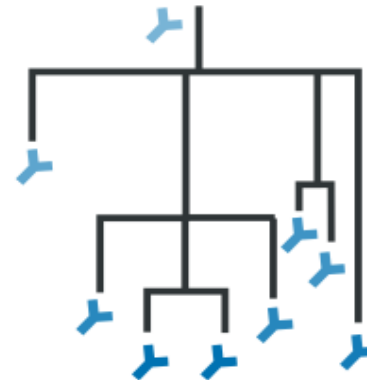


Statistical Problems

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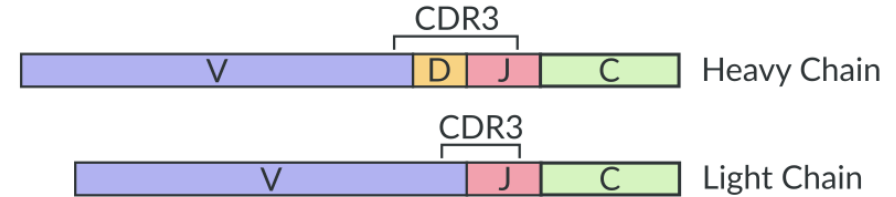


BCRs can mutate

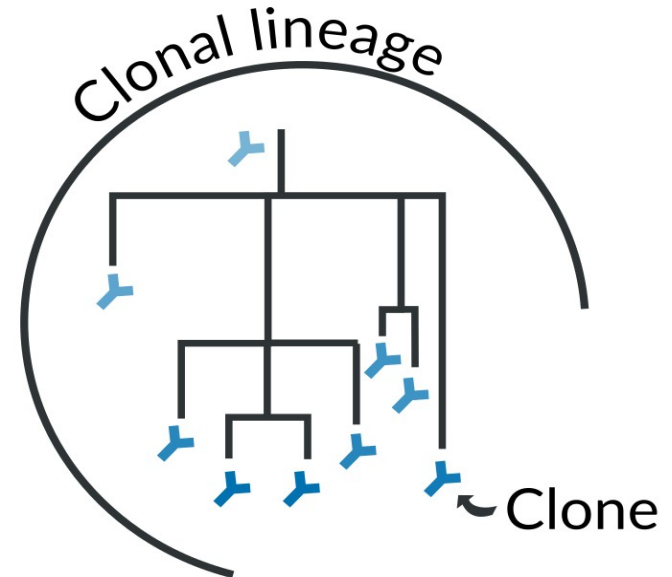


Statistical Problems

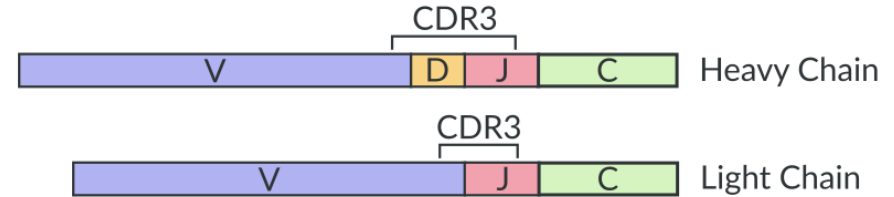
BCRs are randomly generated by V(D)J recombination.



BCRs can mutate

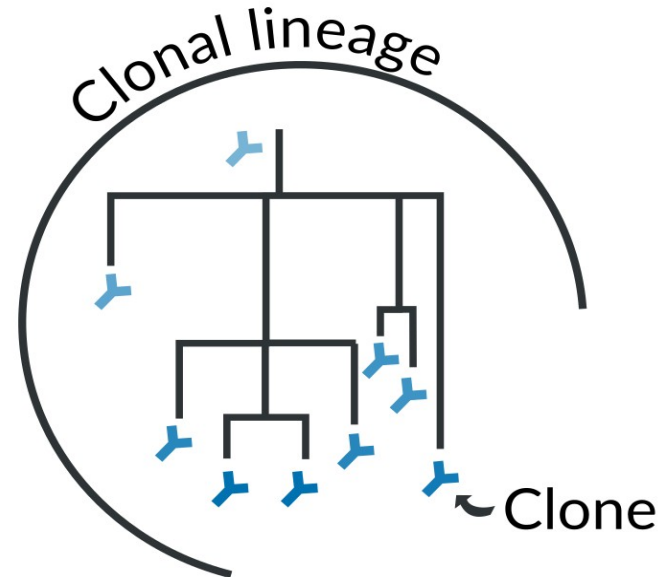


BCRs are randomly generated by V(D)J recombination.

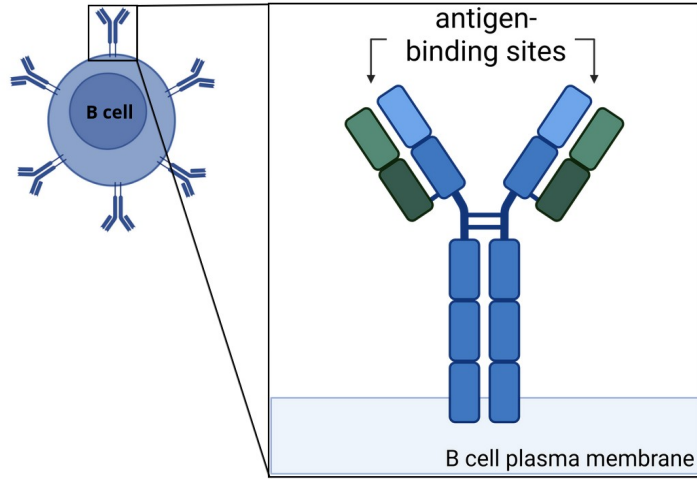


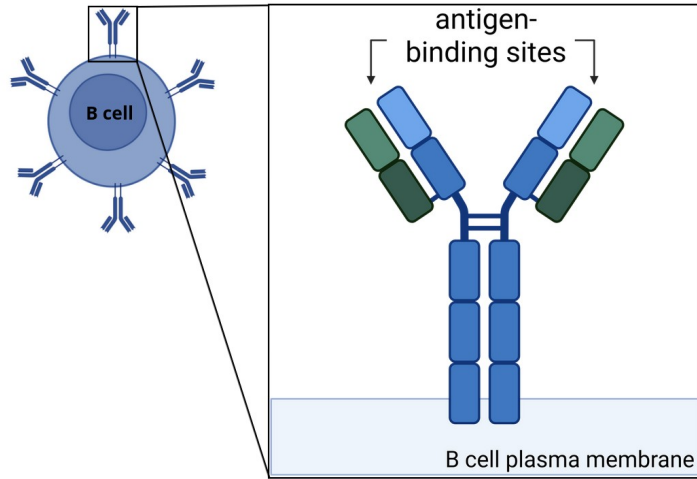
BCRs can mutate

- Each individual repertoire is unique
- Most BCRs are not shared
- Not all BCRs are as likely
- Clonal reconstruction, phylogeny ...

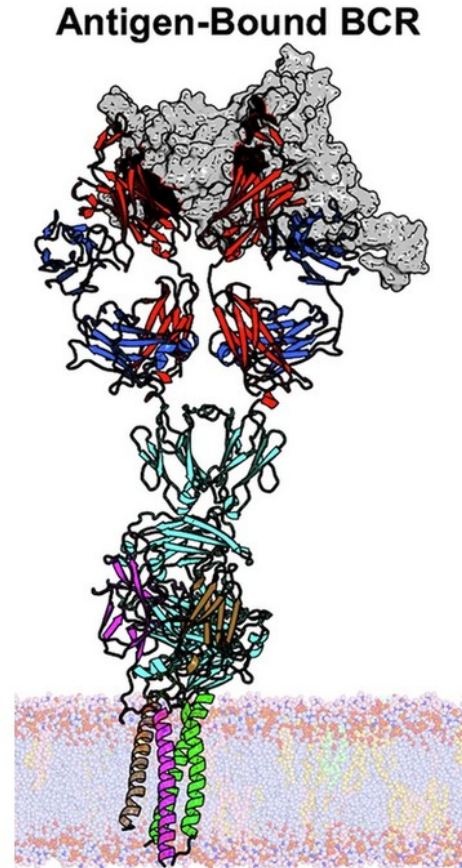


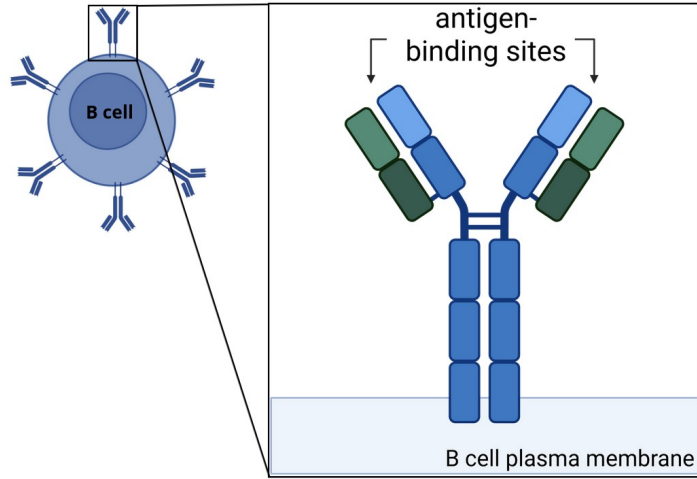
Protein Problems



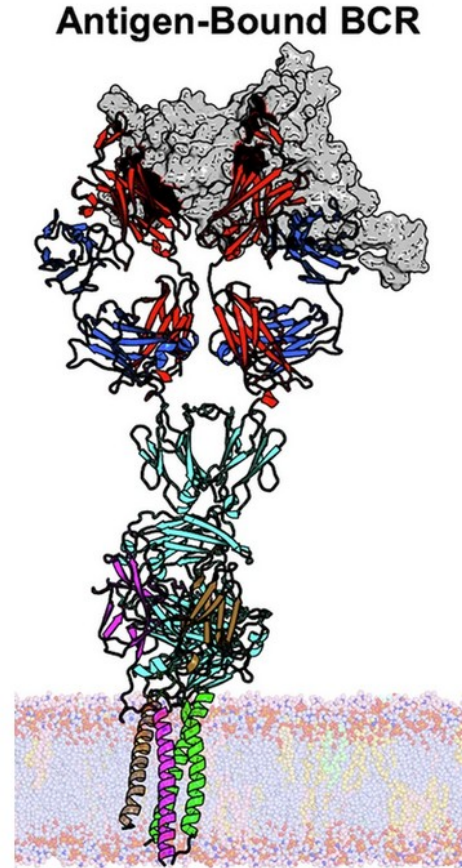


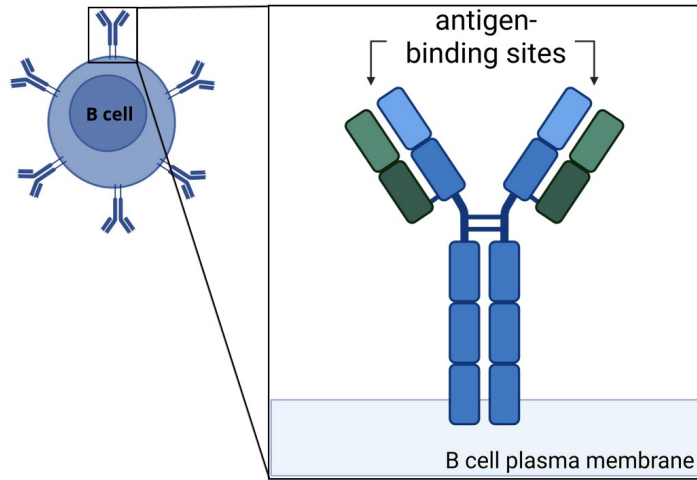
BCR / Antigen interaction



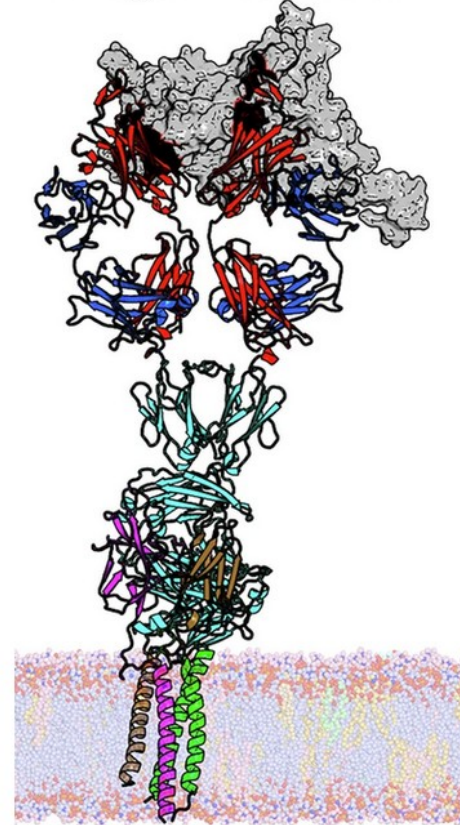


BCR / Antigen interaction
Prediction is near-impossible





Antigen-Bound BCR



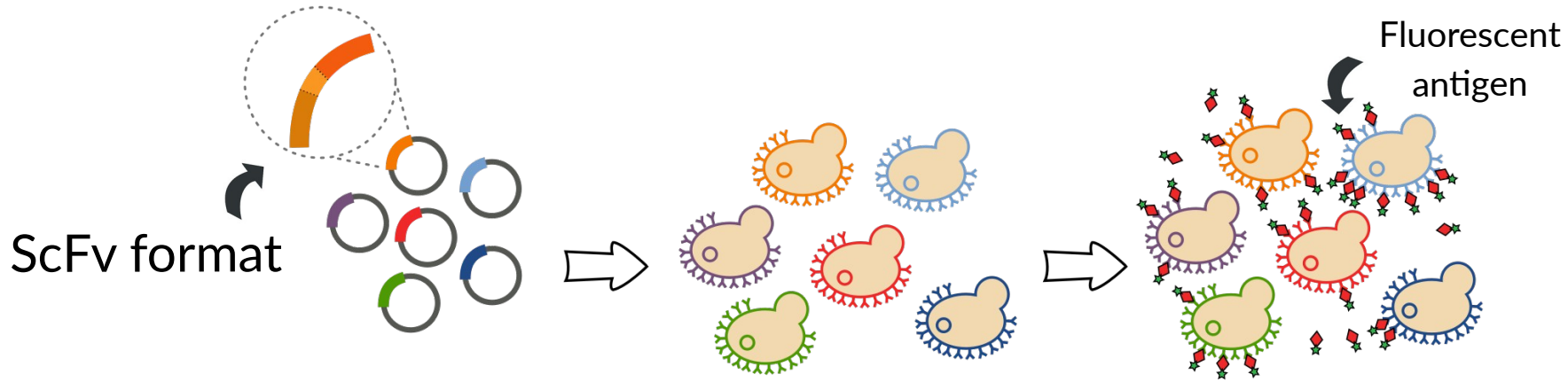
BCR / Antigen interaction

Prediction is near-impossible

Measurement is slow and expensive

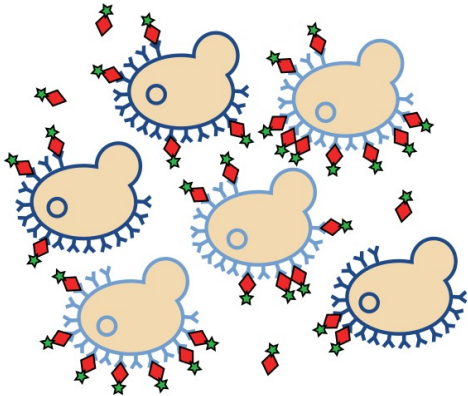
Functional Characterization of the Lineages

Yeast display expression of antibodies



Adams, R. M., et al. eLife (2016)

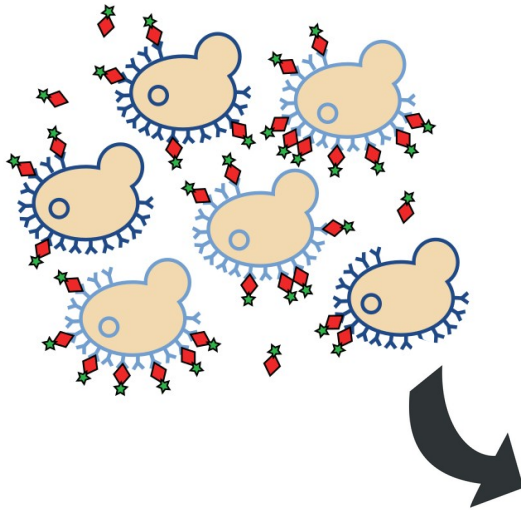
Incubation at a fixed
antigen concentration



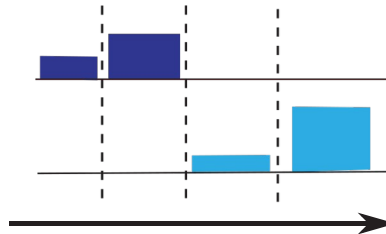
Affinity Measurement in High-Throughput, Tite-Seq

Adams, R. M., et al. eLife (2016)

Incubation at a fixed
antigen concentration



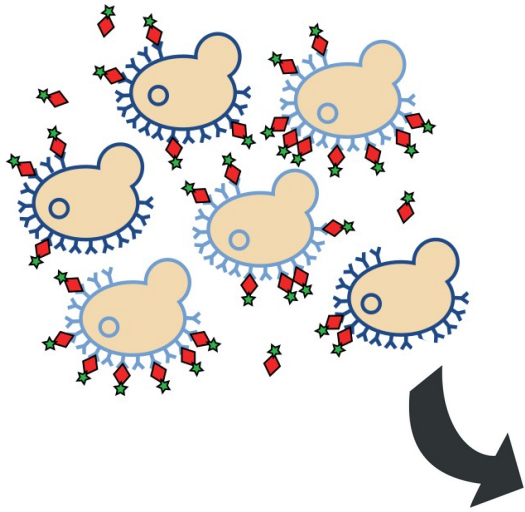
4-ways sort &
sequencing



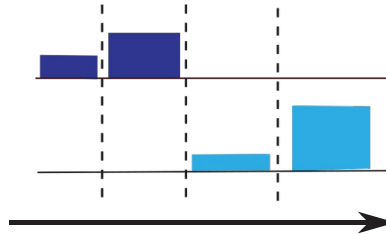
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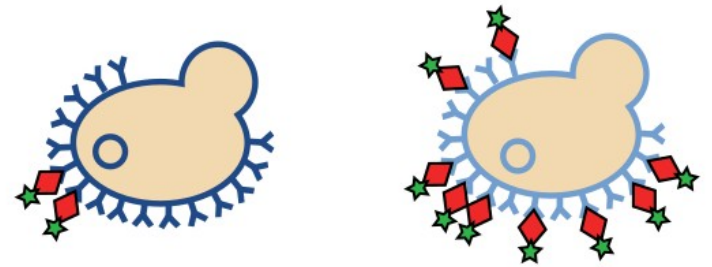
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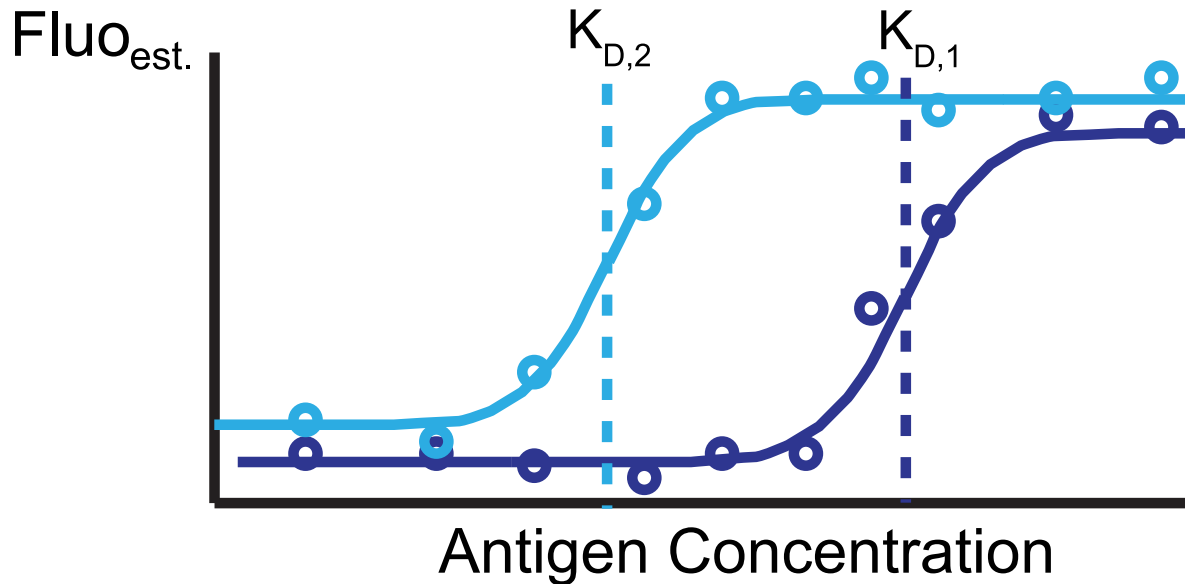


⇒ Estimate the
average proportion of
antigen bound to
each distinct antibody



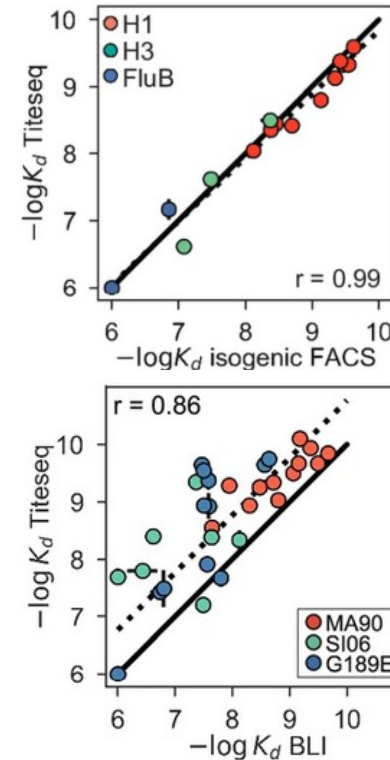
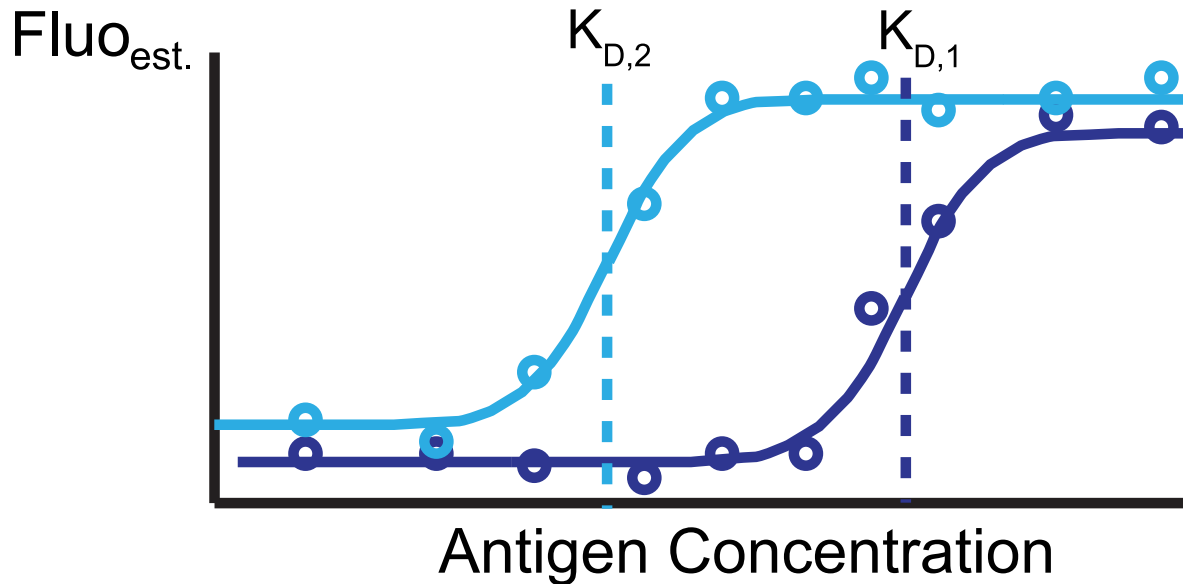
Affinity Measurement in High-Throughput, Tite-Seq

Repeat at different antigen concentrations to create the full titration curve.



Affinity Measurement in High-Throughput, Tite-Seq

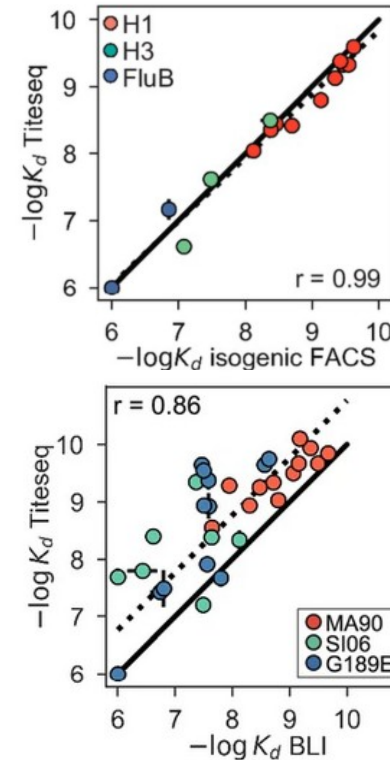
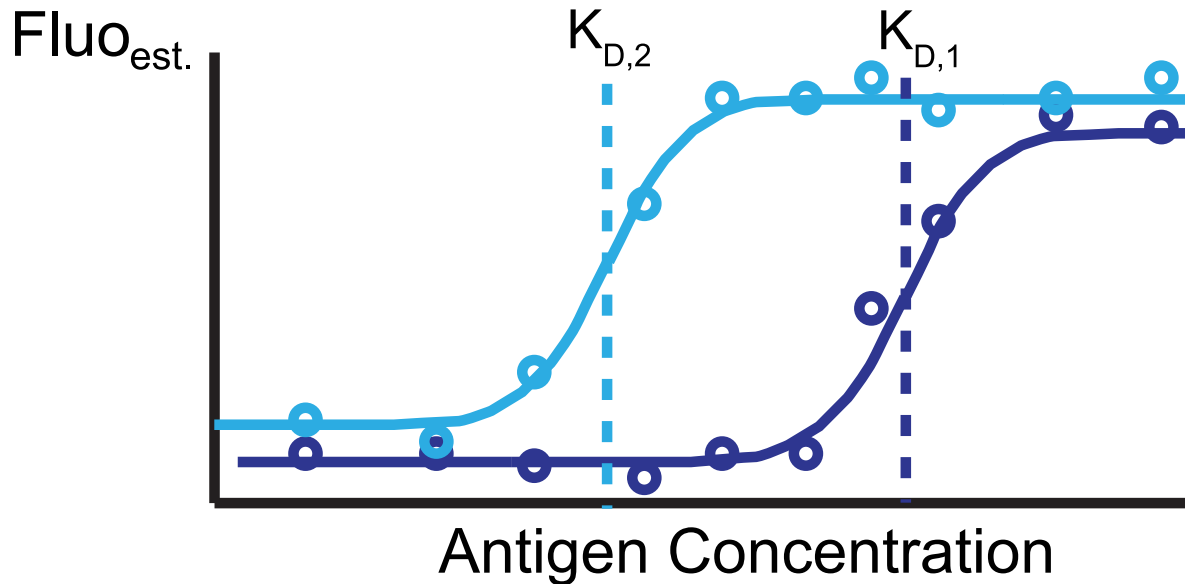
Repeat at different antigen concentrations to create the full titration curve.



Versus titration
interferometry
Versus Bio-Layer

Affinity Measurement in High-Throughput, Tite-Seq

Repeat at different antigen concentrations to create the full titration curve.



Versus titration
interferometry

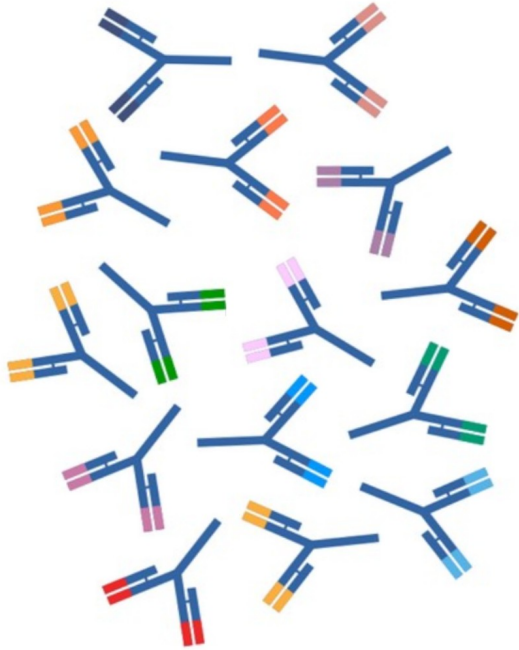
Versus Bio-Layer

Philips, A. M., Lawrence, K. et al., *eLife* (2021)
Moulana, A., Dupic, T. et al., *Nat. Comm.* (2022)
Moulana, A., Dupic, T. et al., *eLife* (2023)...

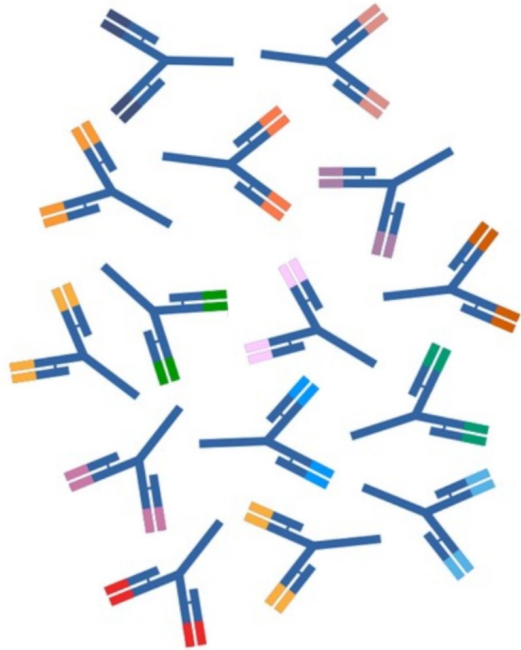
Phillips, A. M. et al, *eLife* (2023)

From one clonal lineage to the repertoire

Real responses are multi-clonal



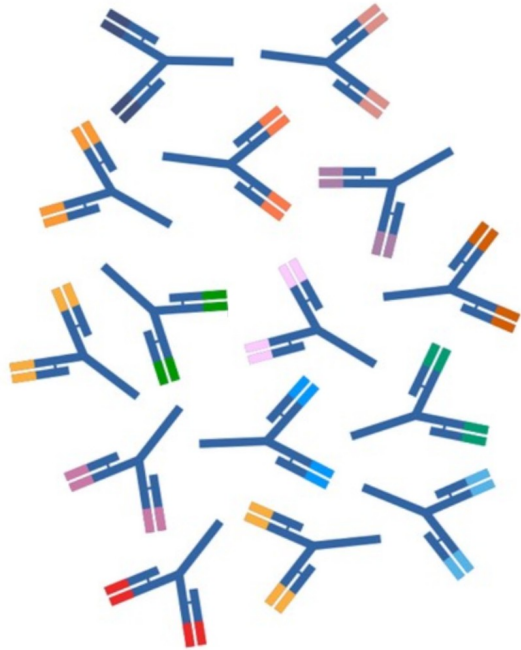
From one clonal lineage to the repertoire



Real responses are multi-clonal

How many lineages are responding to the antigen(s) ?

From one clonal lineage to the repertoire

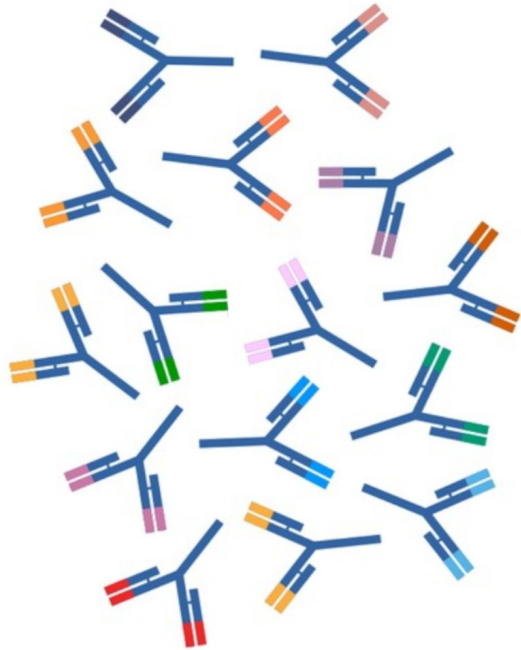


Real responses are multi-clonal

How many lineages are responding
to the antigen(s) ?

What are their properties ?

From one clonal lineage to the repertoire



Real responses are multi-clonal

How many lineages are responding
to the antigen(s) ?

What are their properties ?

Can we detect them computationally ?

Antibody response during Vaccination

Vaccination as a model for studying the repertoire response:
known antigens & known timecourse.

Vaccination as a model for studying the repertoire response:
known antigens & known timecourse.

**Dynamics of B cell repertoires and emergence of
cross-reactive responses in patients with different
severities of COVID-19**

Zachary Montague ¹, Huibin Lv ², Jakub Otwinowski ³, William S DeWitt ⁴, Giulio Isacchini ⁵,
Garrick K Yip ², Wilson W Ng ², Owen Tak-Yin Tsang ⁶, Meng Yuan ⁷, Hejun Liu ⁷, Ian A Wilson ⁸,
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Convergent antibody evolution and clonotype expansion following influenza virus vaccination

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B cell receptor repertoire kinetics after SARS-CoV-2 infection and vaccination

Prasanti Kotagiri¹, Federica Mescia², William M Rae², Laura Bergamaschi², Zewen K Tuong³, Lorinda Turner², Kelvin Hunter², Pehuén P Gerber², Myra Hosmillo⁴;

Cambridge Institute of Therapeutic Immunology and Infectious Disease-National Institute of Health Research (CITIID-NIHR) COVID BioResource Collaboration;

Christoph Hess⁵, Menna R Clatworthy⁶, Ian G Goodfellow⁴, Nicholas J Matheson⁷, Eoin F McKinney², Mark R Wills², Ravindra K Gupta², John R Bradley⁸, Rachael J M Bashford-Rogers⁹, Paul A Lyons¹⁰, Kenneth G C Smith¹¹

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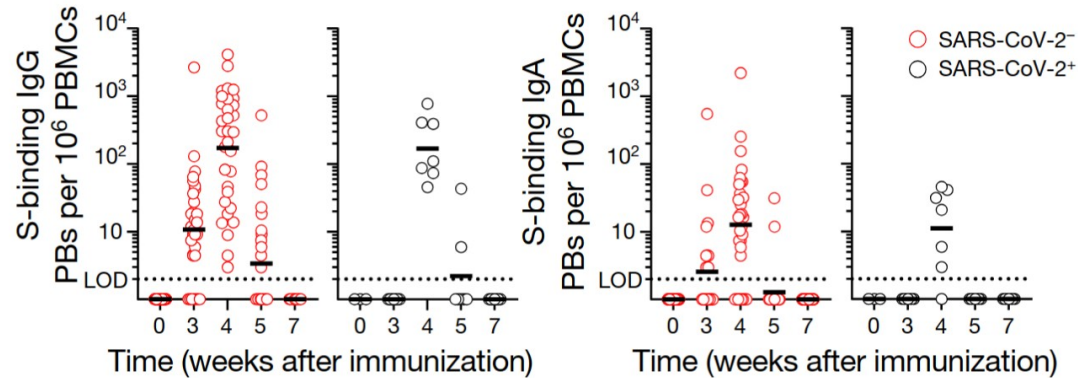
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Longitudinal analysis of the peripheral B cell repertoire reveals unique effects of immunization with a new influenza virus strain

Bernardo Cortina-Ceballos¹, Elizabeth Ernestina Godoy-Lozano¹, Juan Téllez-Sosa¹, Marbella Ovilla-Muñoz¹, Hugo Sámano-Sánchez¹, Andrés Aguilar-Salgado¹, Rosa Elena Gómez-Barreto¹, Humberto Valdovinos-Torres¹, Irma López-Martínez², Rodrigo Aparicio-Antonio², Mario H Rodríguez¹, Jesús Martínez-Barnetche³

PBMC B-cell response to SARS-CoV-2 vaccination

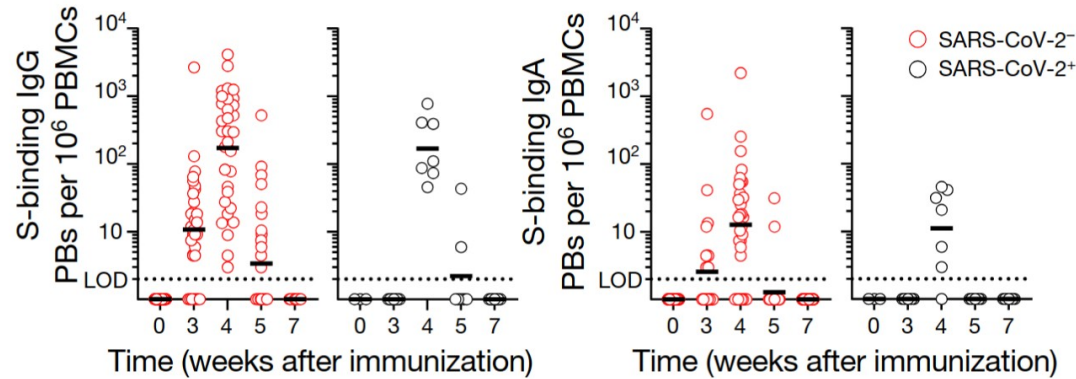
PBMC B-cell response to SARS-CoV-2 vaccination



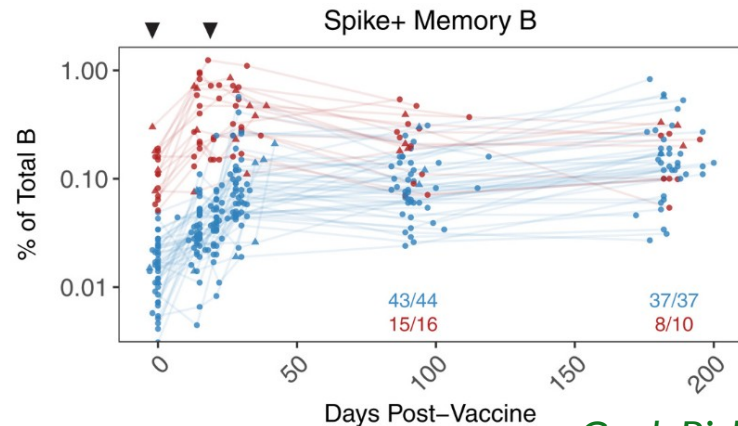
Short term (< 6 weeks):
Plasmablast and antibody
production

Turner, J. S. et al. Nature (2021)

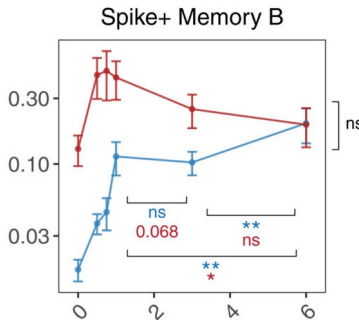
PBMC B-cell response to SARS-CoV-2 vaccination



Turner, J. S. et al. Nature (2021)



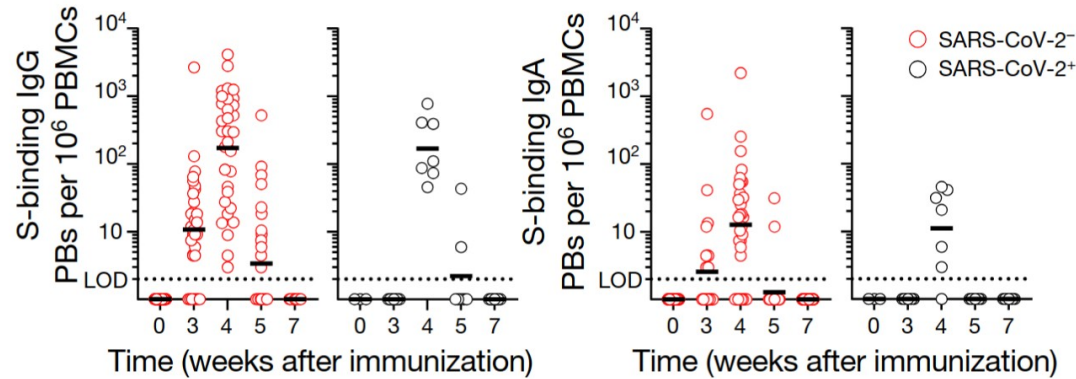
Goel, Rishi R. et al. Science (2021)



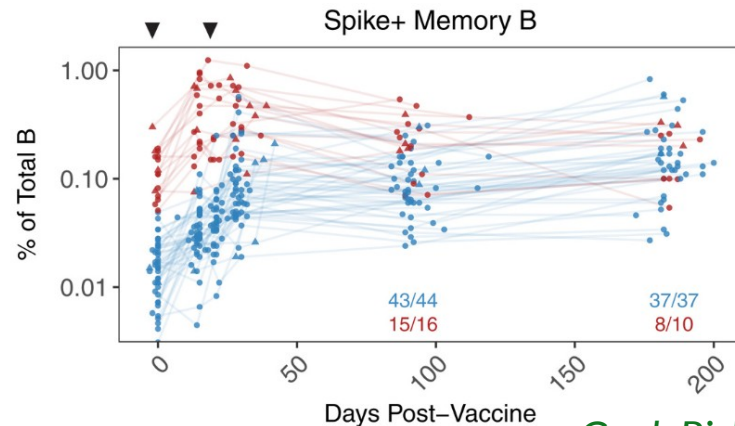
Short term (< 6 weeks):
Plasmablast and antibody
production

Long term:
Memory B-cells

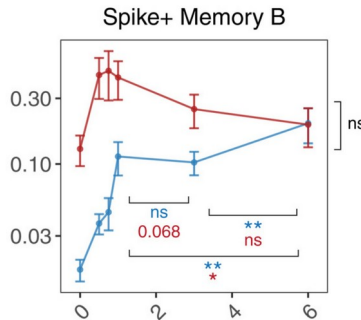
PBMC B-cell response to SARS-CoV-2 vaccination



Turner, J. S. et al. Nature (2021)



Goel, Rishi R. et al. Science (2021)



Short term (< 6 weeks):
Plasmablast and antibody
production

Long term:
Memory B-cells

0.1-1% of the total B-
cells are spike+

Antibody response during Vaccination

Following the evolution of the BCR repertoire alongside SARS-CoV-2 vaccination.

Antibody response during Vaccination

Following the evolution of the BCR repertoire alongside SARS-CoV-2 vaccination.

1. Long-term response

Antibody response during Vaccination

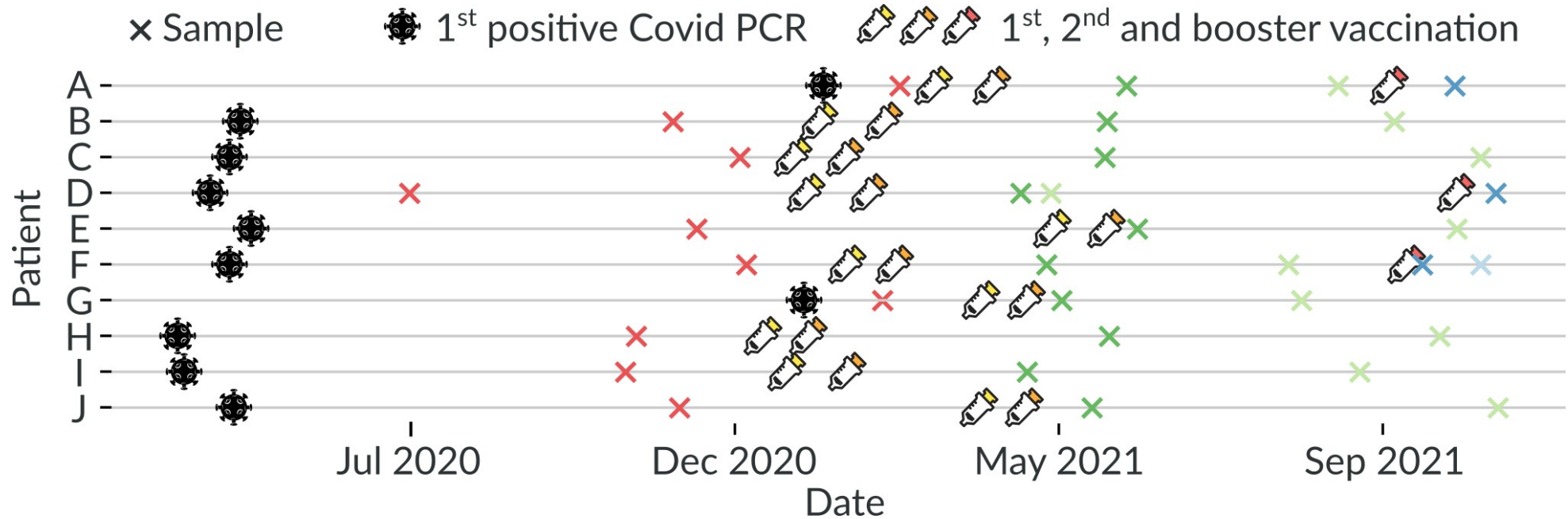
Following the evolution of the BCR repertoire alongside SARS-CoV-2 vaccination.

1. Long-term response
2. High depth

Following the evolution of the BCR repertoire alongside SARS-CoV-2 vaccination.

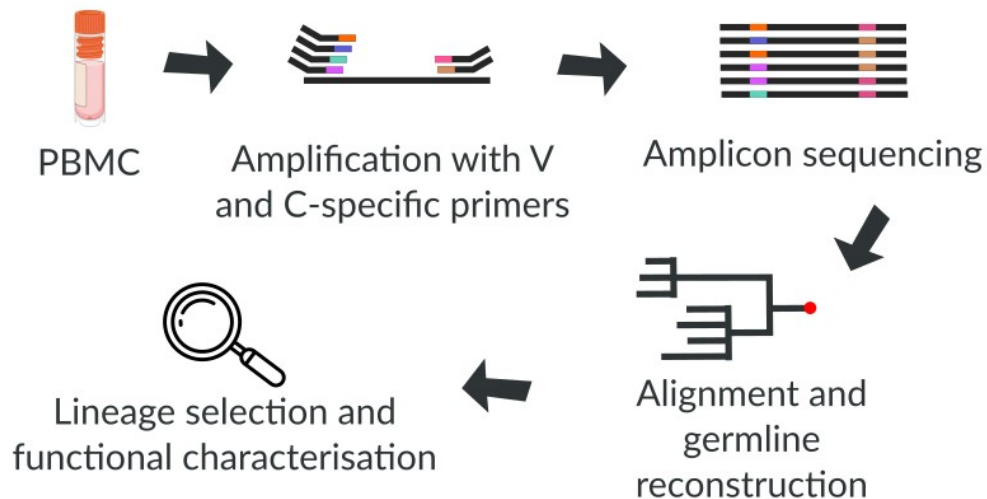
1. Long-term response
2. High depth
3. Functional characterization

Samples timeline



Longitudinal sampling on 10 individuals. Collaboration with the Yu lab at the Ragon institute, 10^9 B cells.

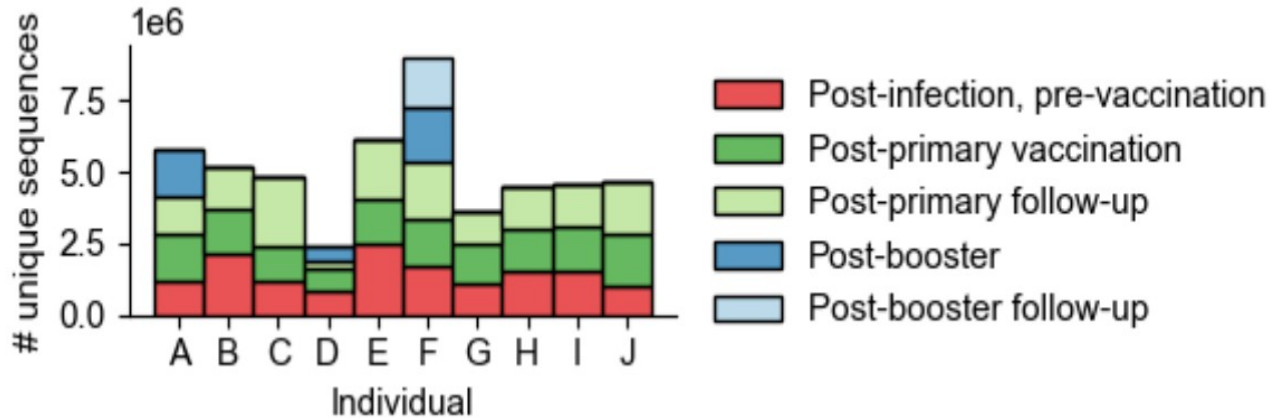
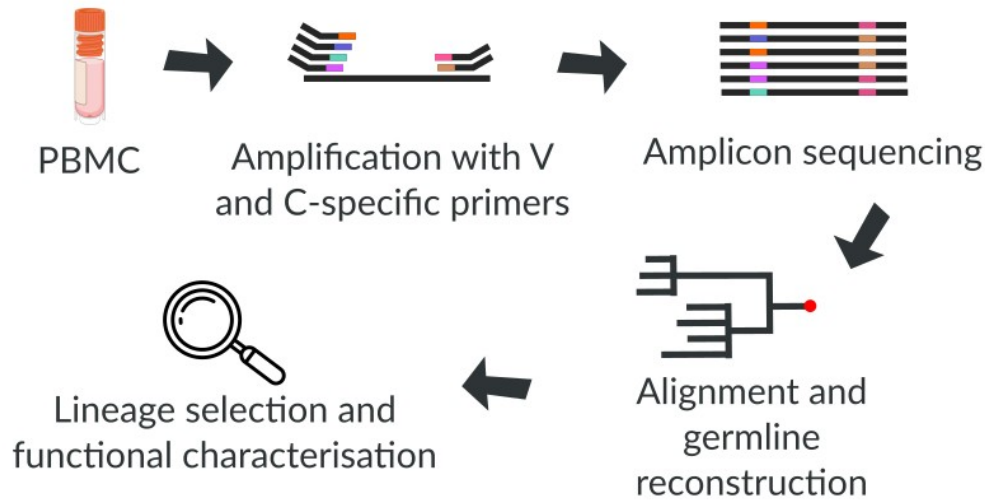
Sequencing



Deep bulk repertoire sequencing of heavy and light chains, 34 samples $\times$ 5 replicates.

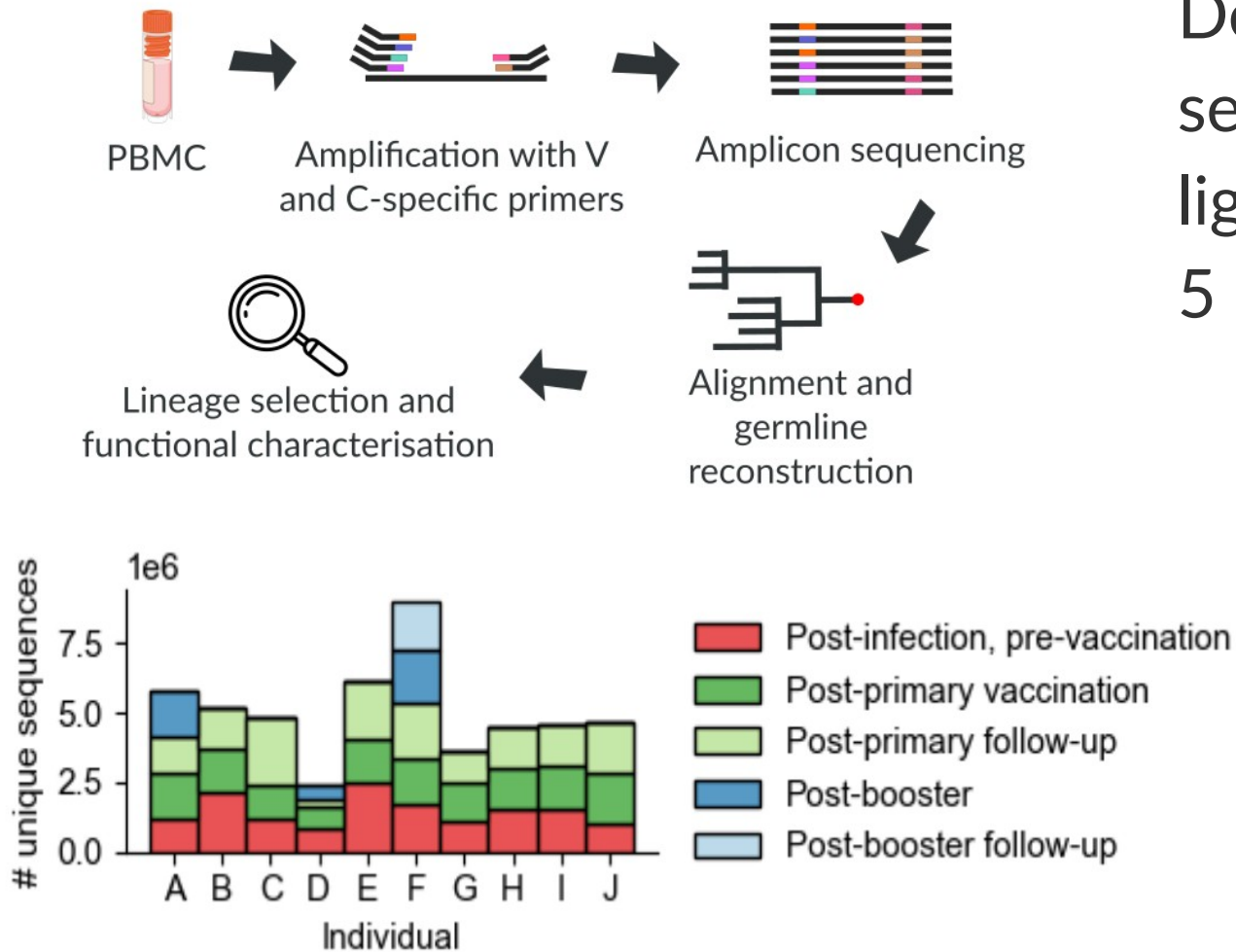
Sequencing

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Sequencing

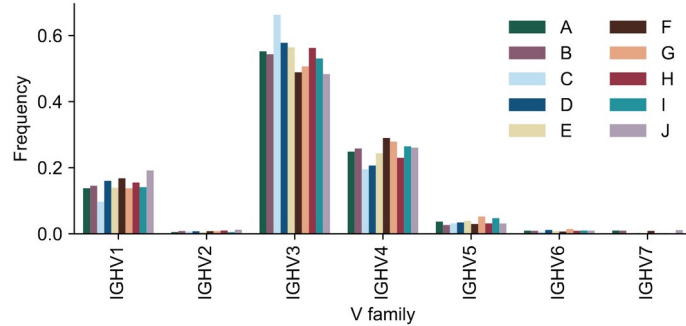
Deep bulk repertoire sequencing of heavy and light chains, 34 samples × 5 replicates.



The stack:

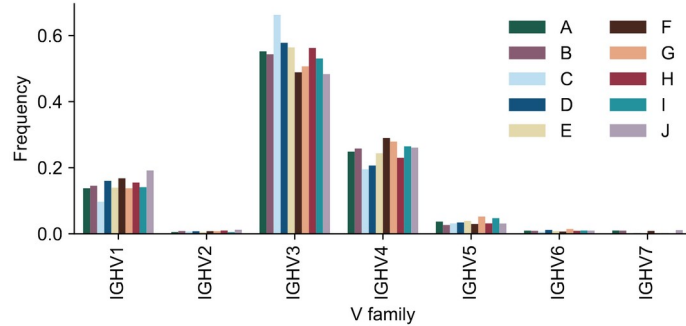
- MiXCR
- Hilary for clonal reconstruction
- IGoR for recombination model
- Igonometree for tree reconstruction

The Immune Repertoire

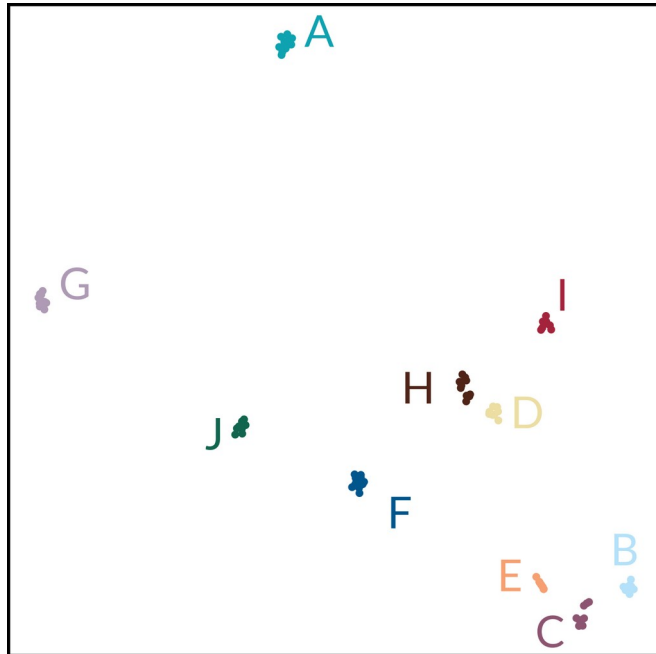


Repertoire change a lot but
their statistical property
don't.

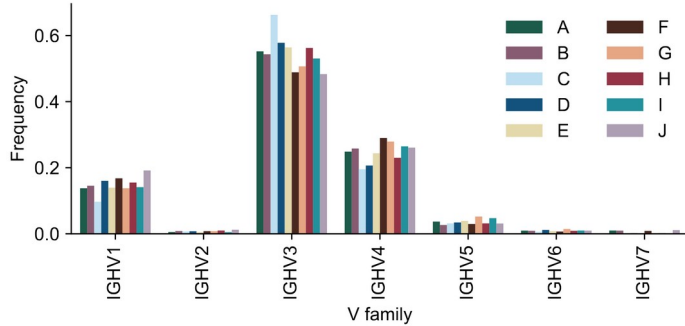
The Immune Repertoire



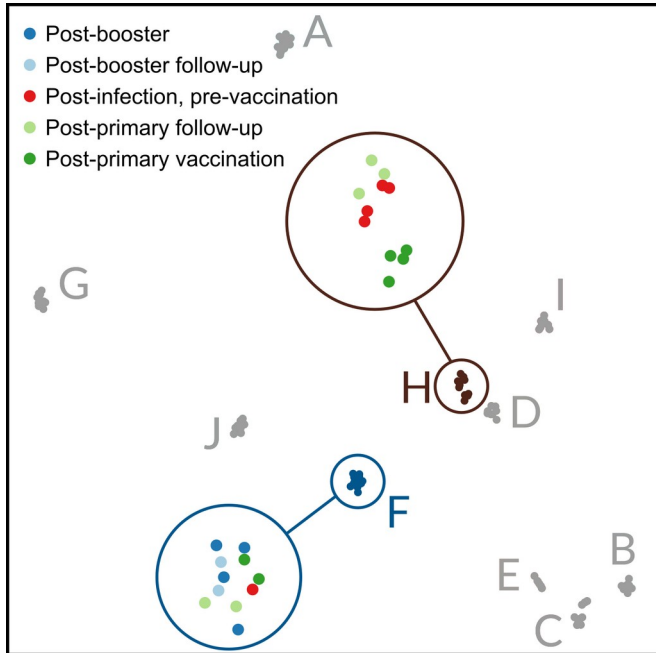
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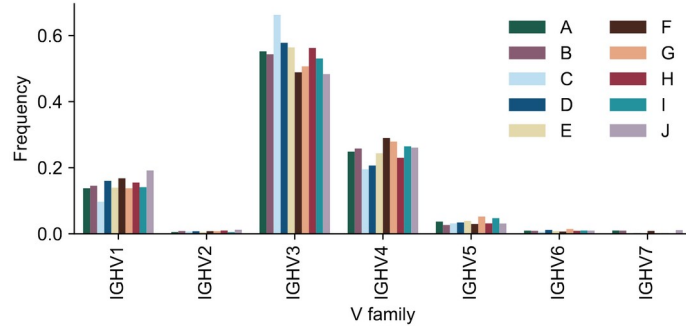
The Immune Repertoire



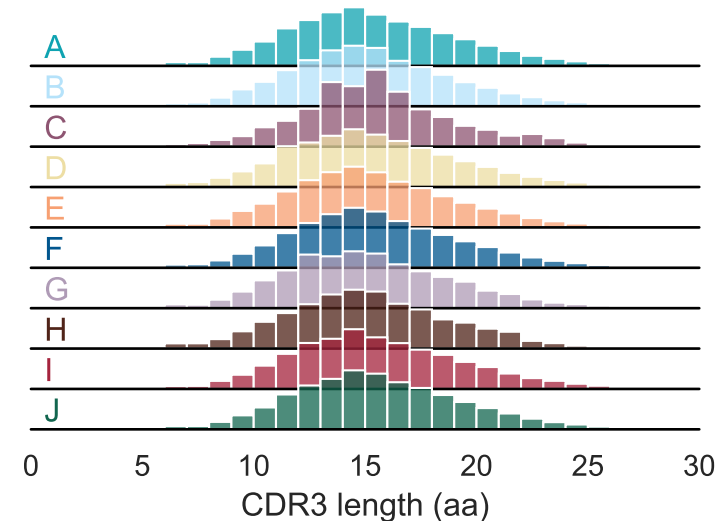
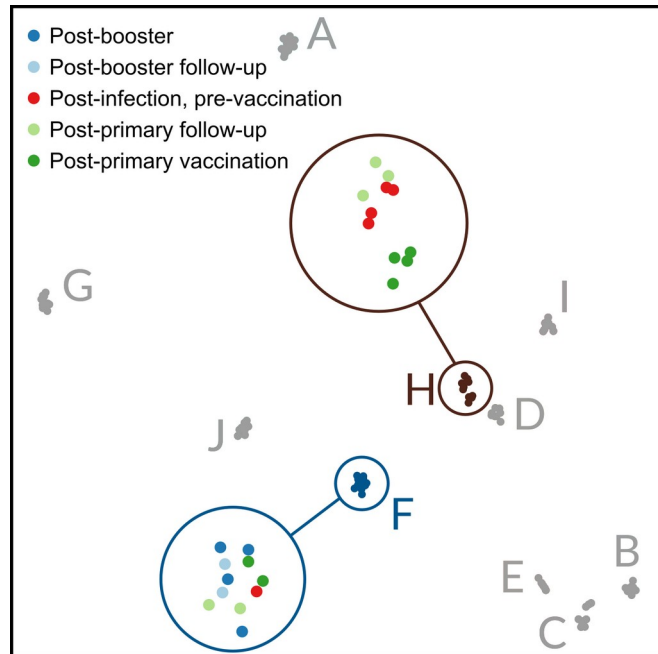
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The Immune Repertoire

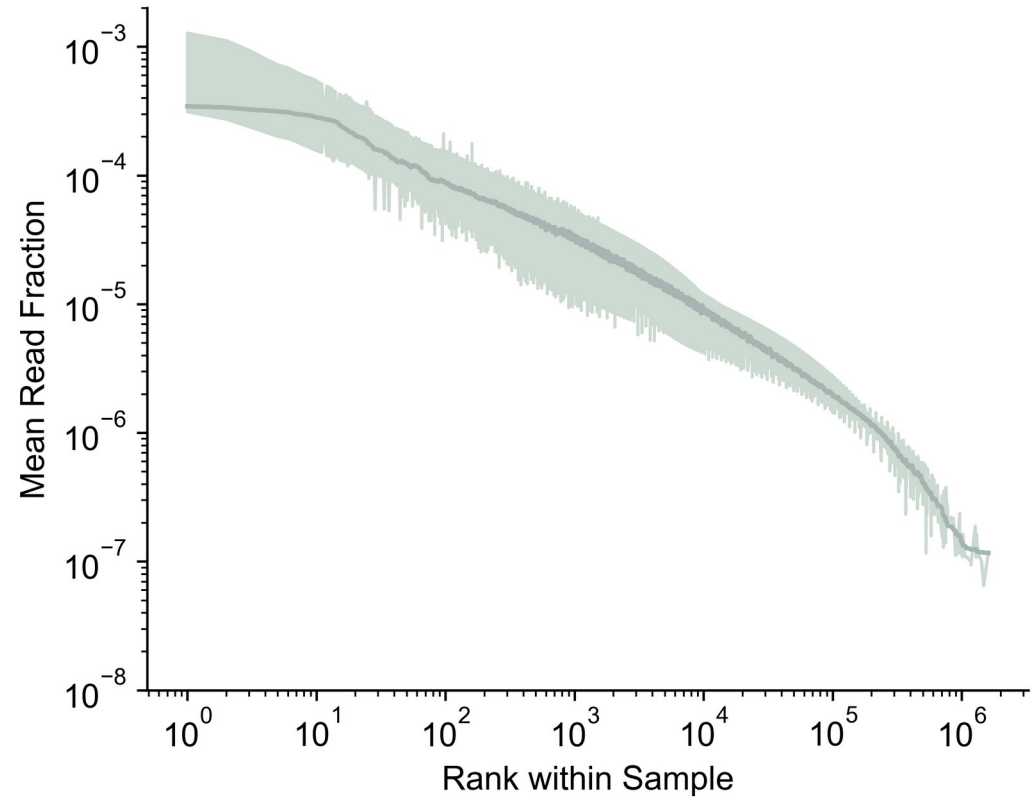


Repertoire change a lot but their statistical property don't.



Making a library

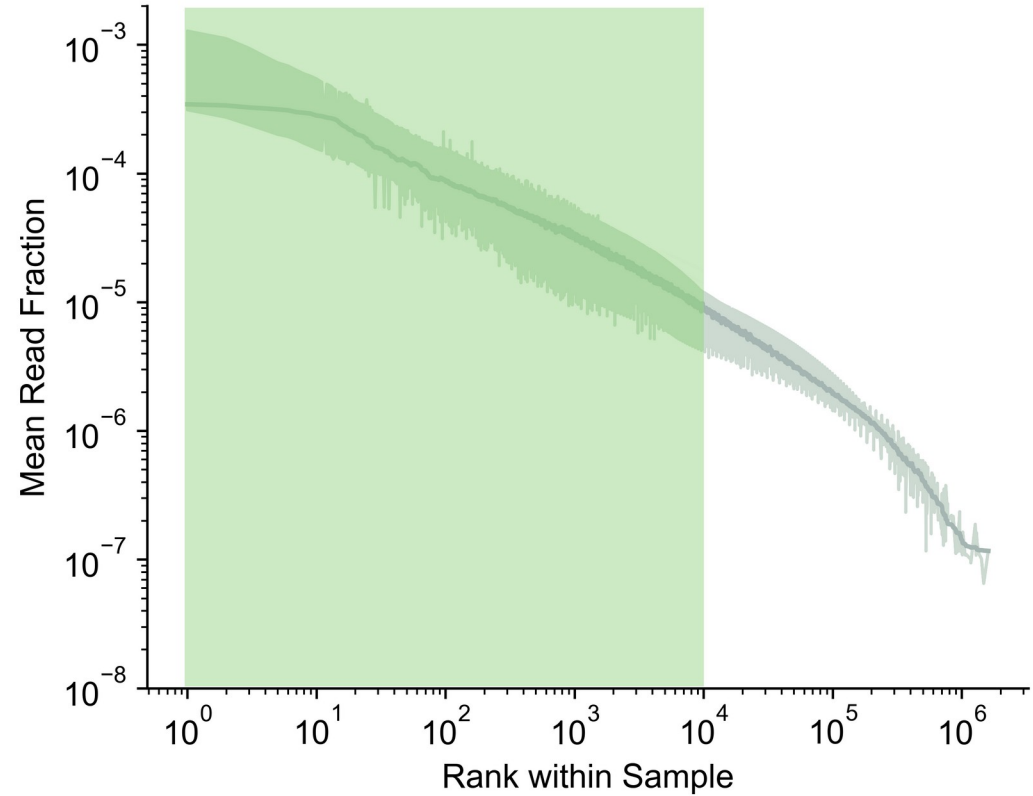
BCR frequencies follow
“power-law” distribution



Making a library

BCR frequencies follow
“power-law” distribution

We focus on the **10'000**
largest clonal lineages.

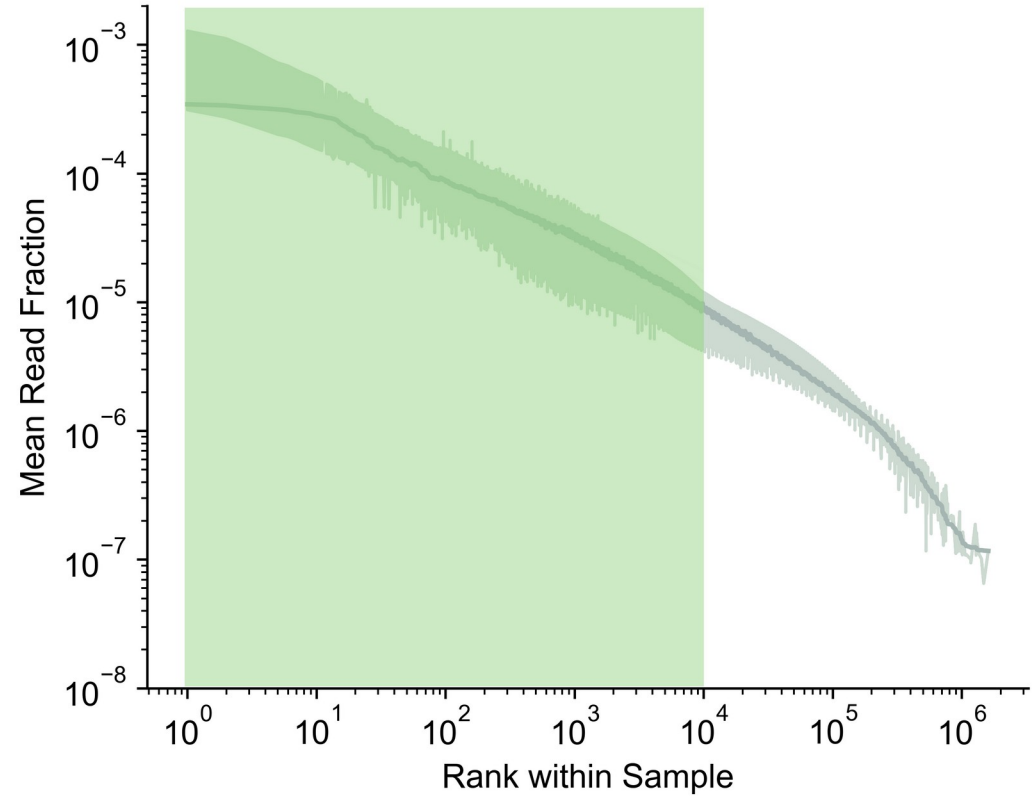


Making a library

BCR frequencies follow
“power-law” distribution

We focus on the **10'000**
largest clonal lineages.

(still a very small part of the overall
repertoire ~ 0.1%)



Making a library

Bulk sequencing limitation: **no heavy-light pairing**

Making a library

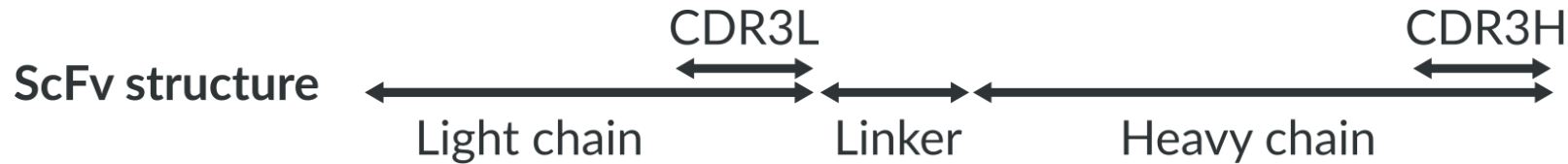
Bulk sequencing limitation: **no heavy-light pairing**

⇒ Combinatorial approach

Making a library

Bulk sequencing limitation: **no heavy-light pairing**

⇒ Combinatorial approach



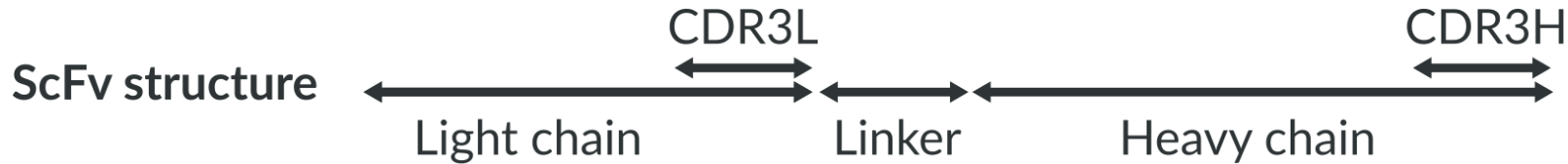
Golden gate assembly



Making a library

Bulk sequencing limitation: **no heavy-light pairing**
 ⇒ Combinatorial approach

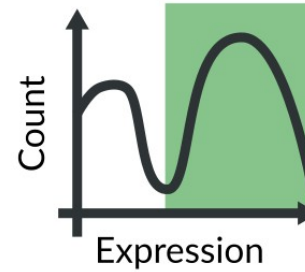
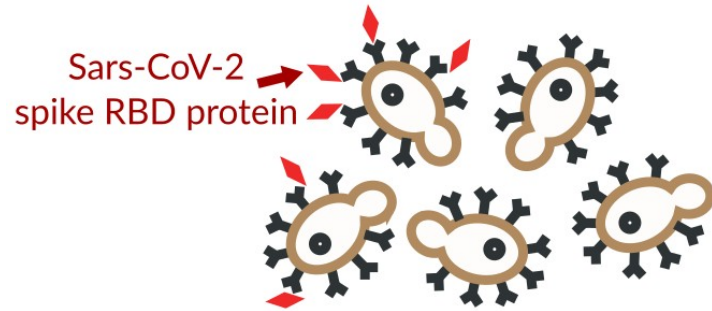
Focus on the **reconstructed naive sequence** →



Golden gate assembly



Functional characterization



Sort for
expression
and **binder**
enrichment



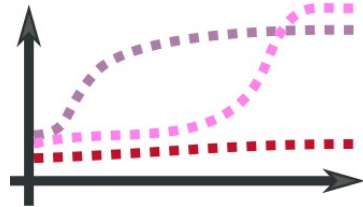
Barcode-scFv association



+ Nanopore sequencing

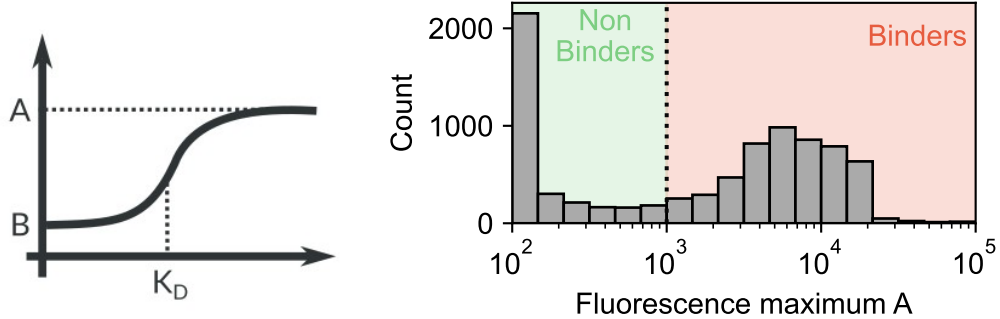
Library size = 68'000

Tite-Seq
High-throughput
 K_D measurements



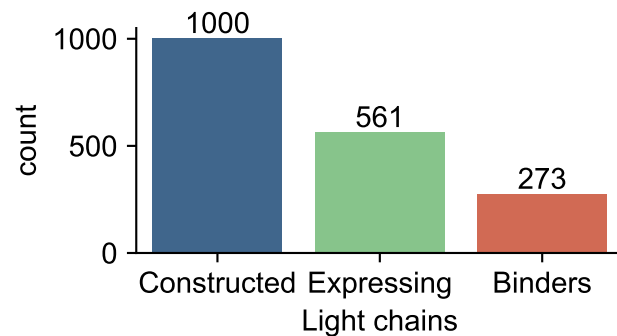
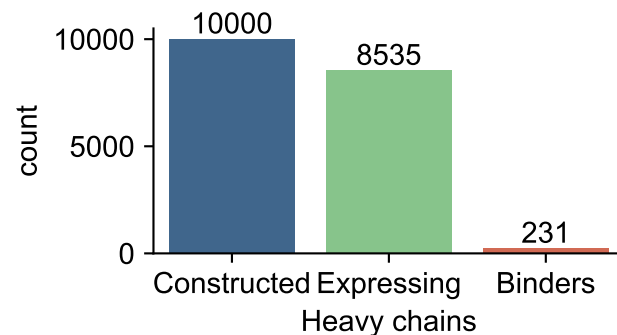
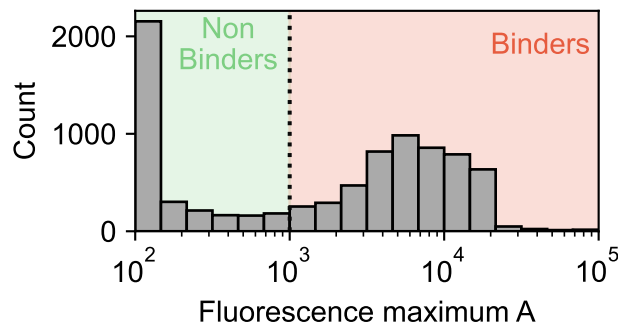
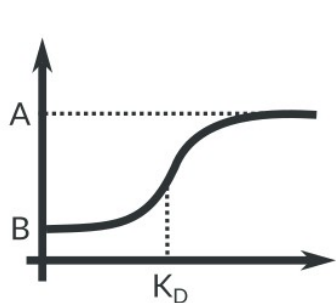
Library size = 10'000

Binders



Tite-Seq allow us to differentiate cleanly binders from non-binders

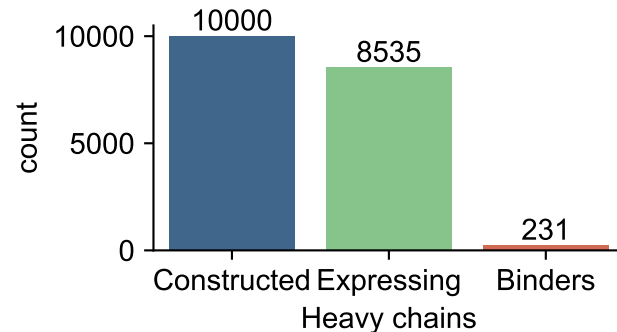
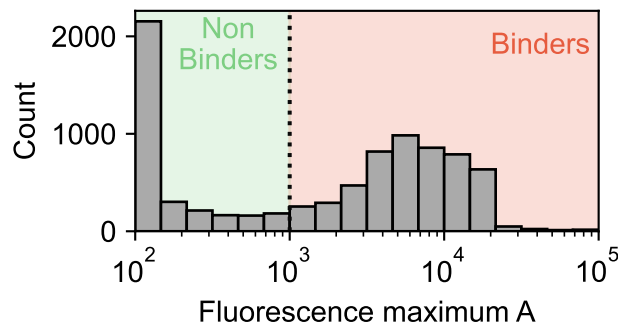
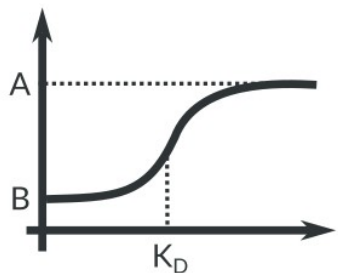
Binders



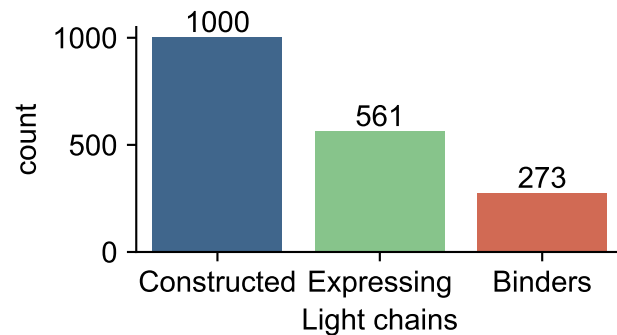
Tite-Seq allow us to differentiate cleanly binders from non-binders

2% of the heavy repertoire binds !

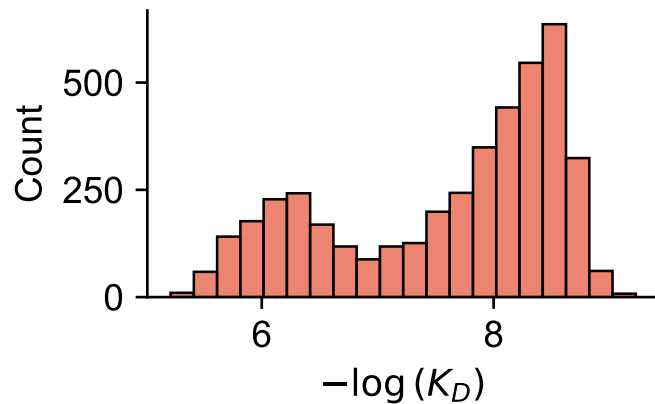
Binders



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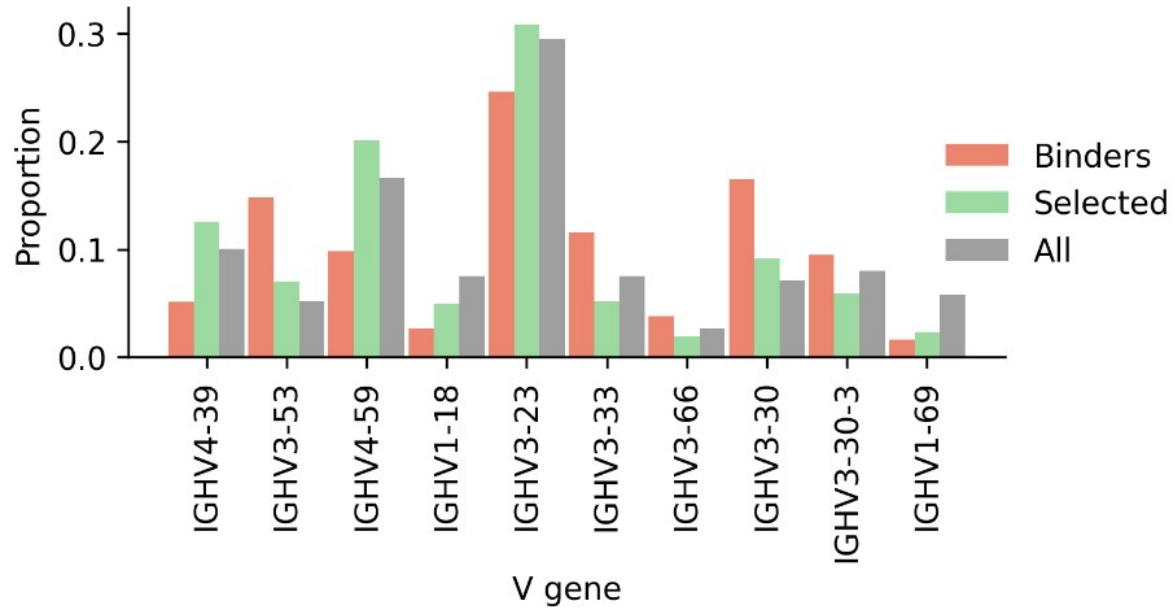


Measured K_D :

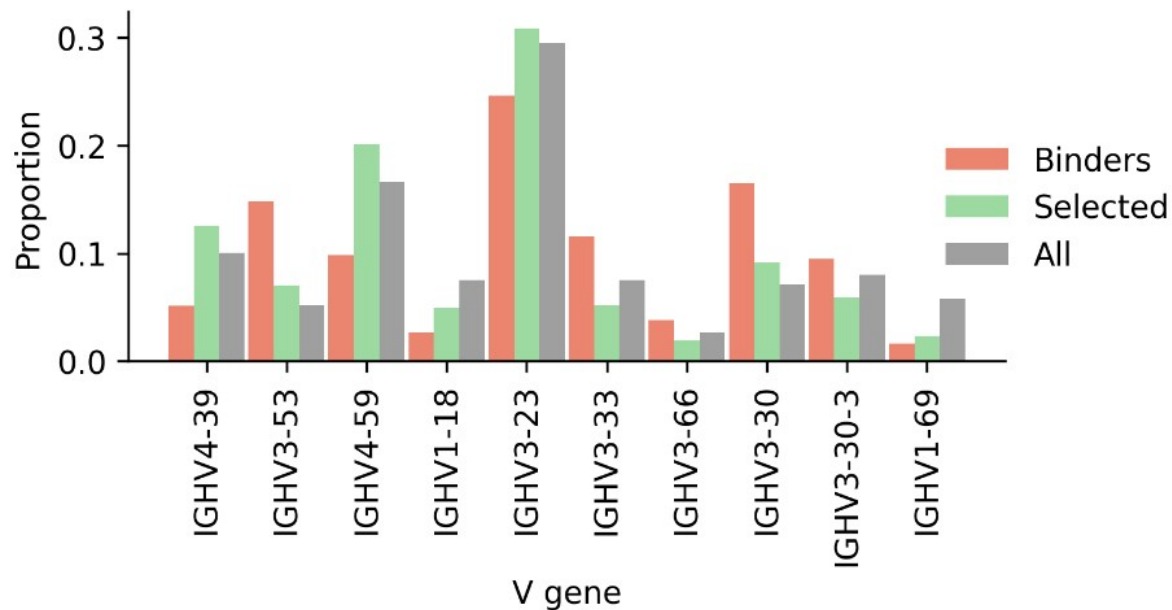


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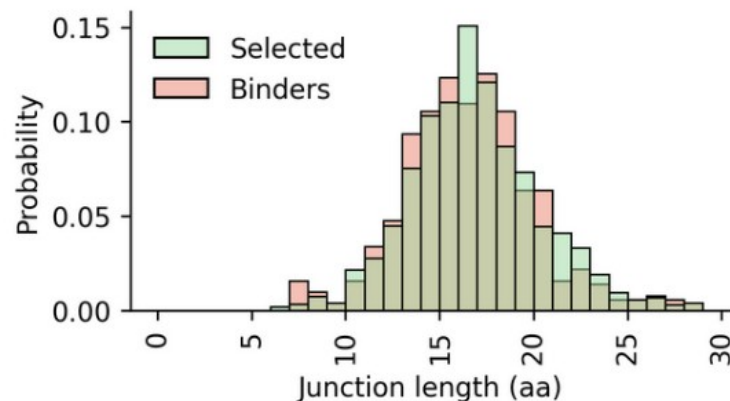
A diverse response



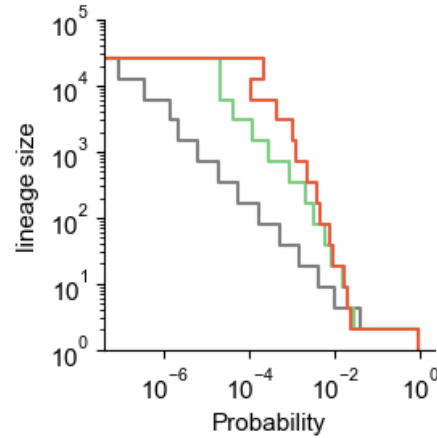
A diverse response



Binder lineages come from all V-genes (that we tested) and have diverse CDR3 lengths

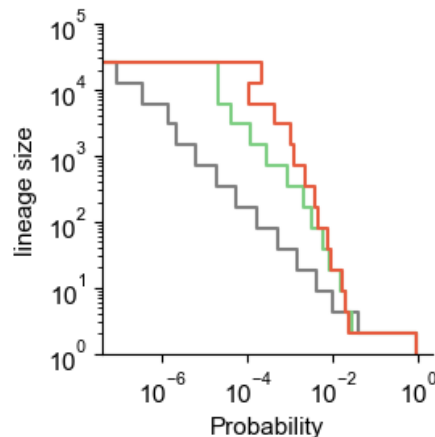


Properties of the “binding” lineages



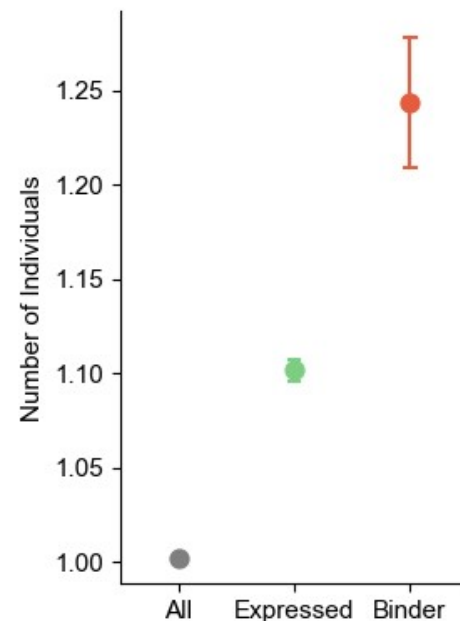
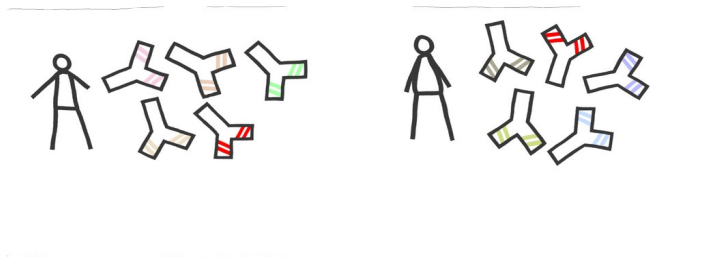
Binder lineages are larger

Properties of the “binding” lineages



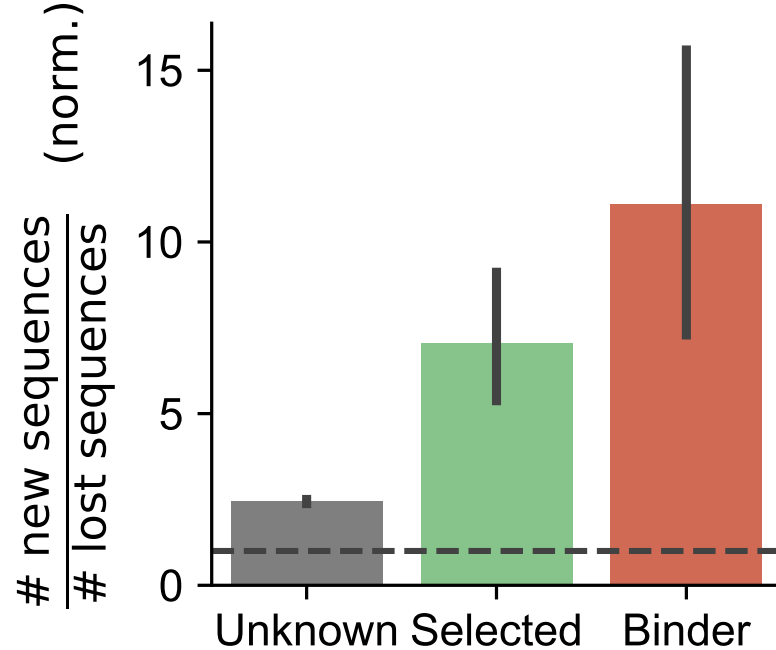
Binder lineages are larger

Binder lineages are more likely to be “public” among vaccinated individuals



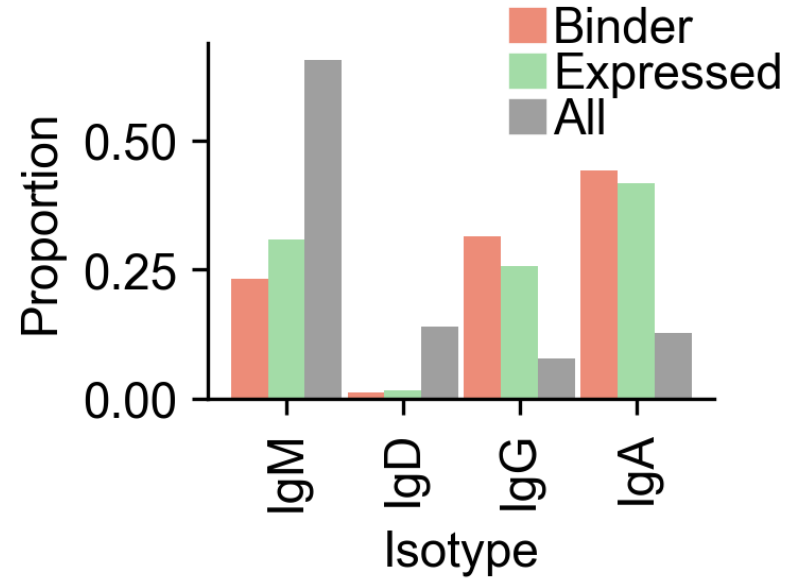
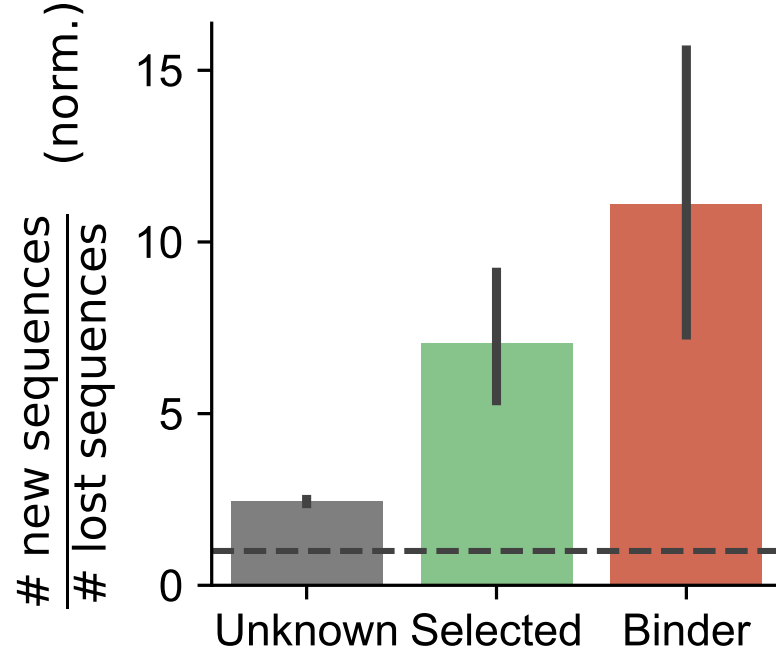
Properties of the “binding” lineages

Lineage with binding germlines
evolve more



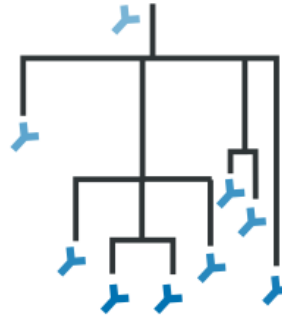
Properties of the “binding” lineages

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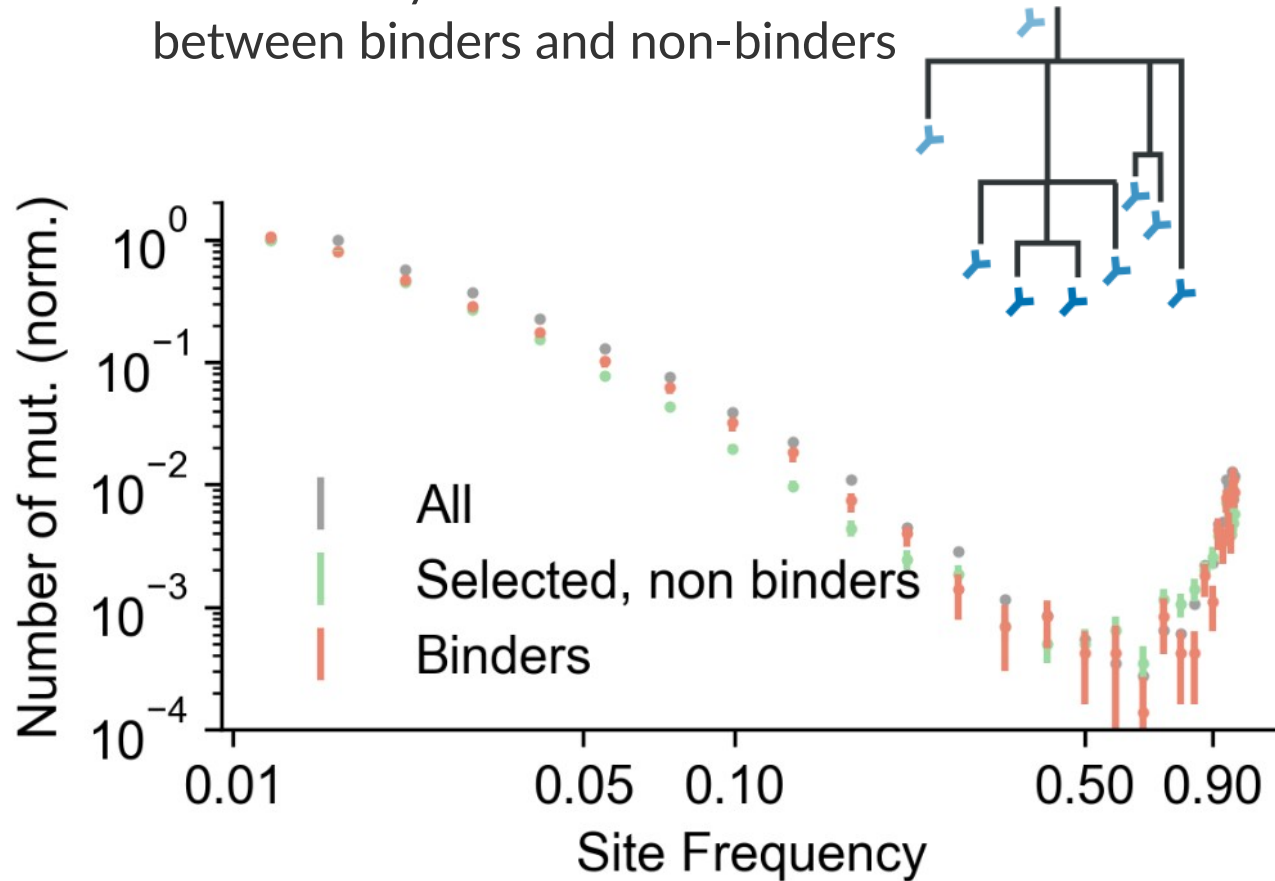
Binder lineages are more
often class-switched

Binder lineages evolve differently



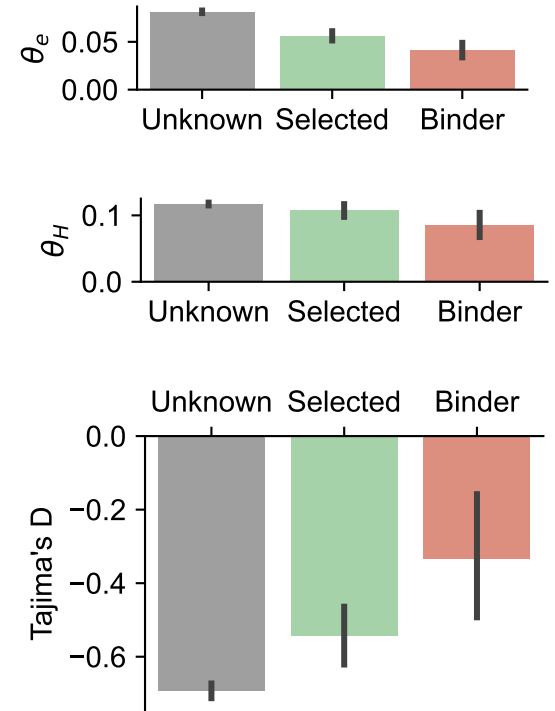
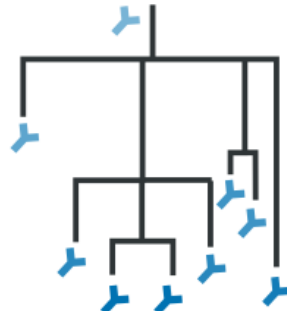
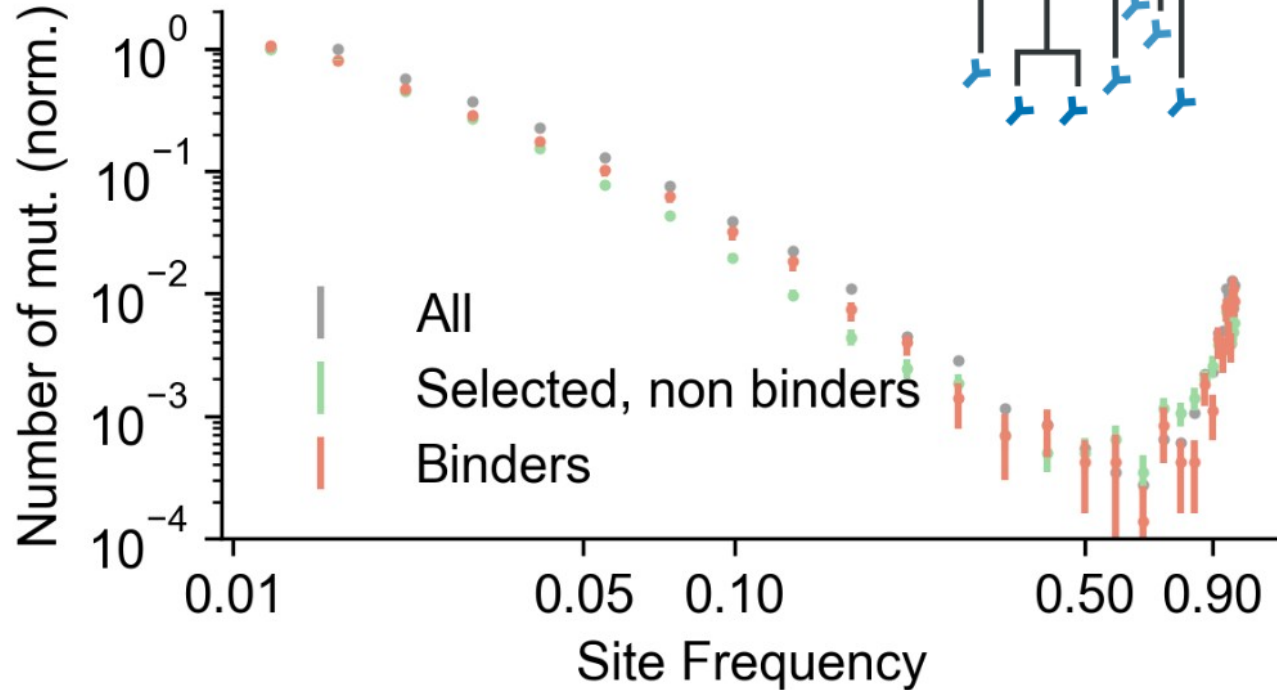
Binder lineages evolve differently

Evolutionary metrics differ
between binders and non-binders



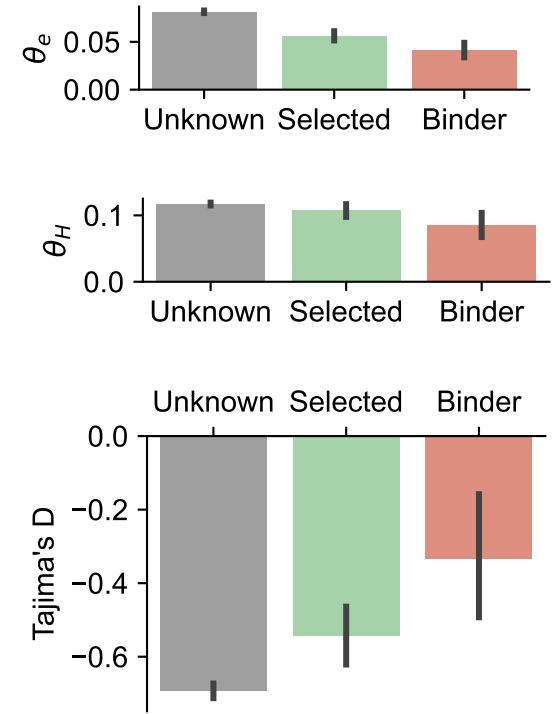
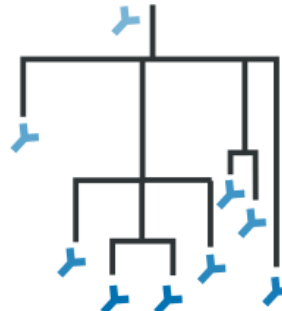
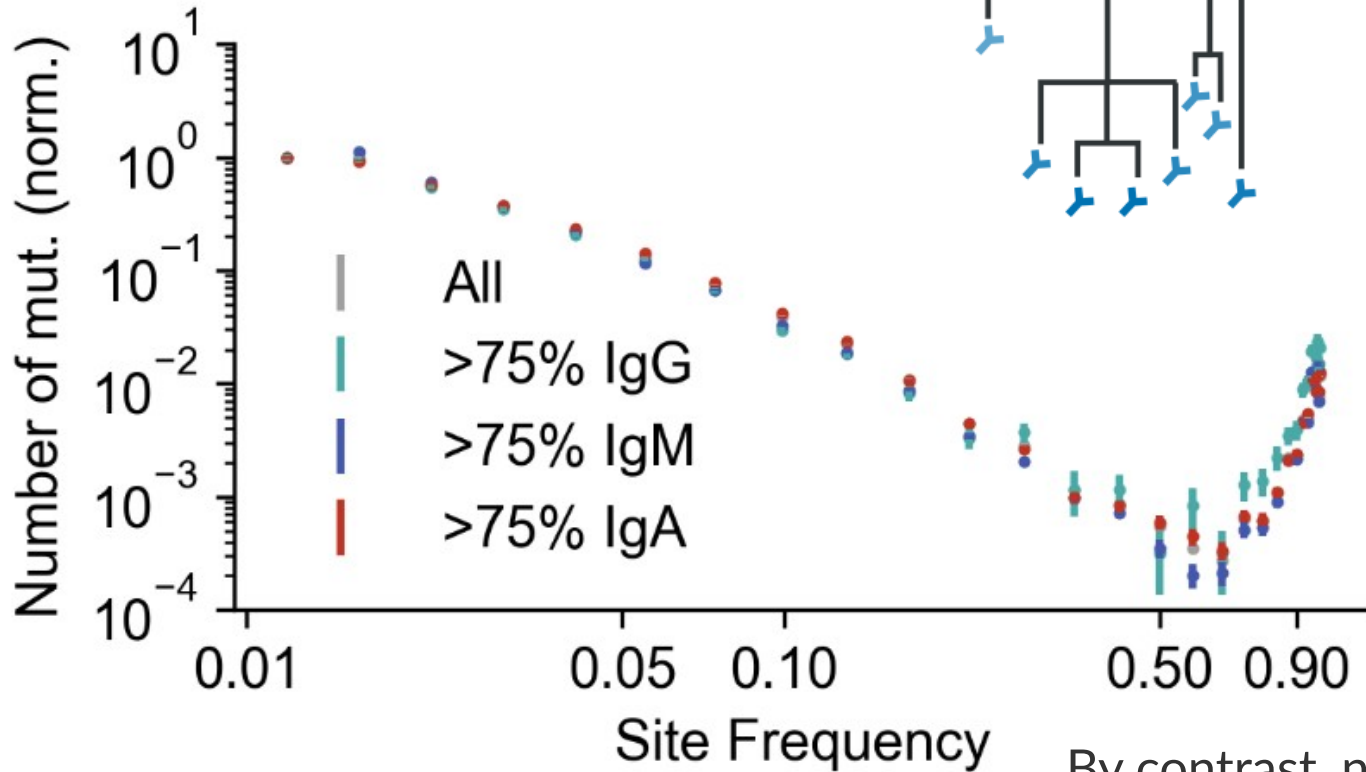
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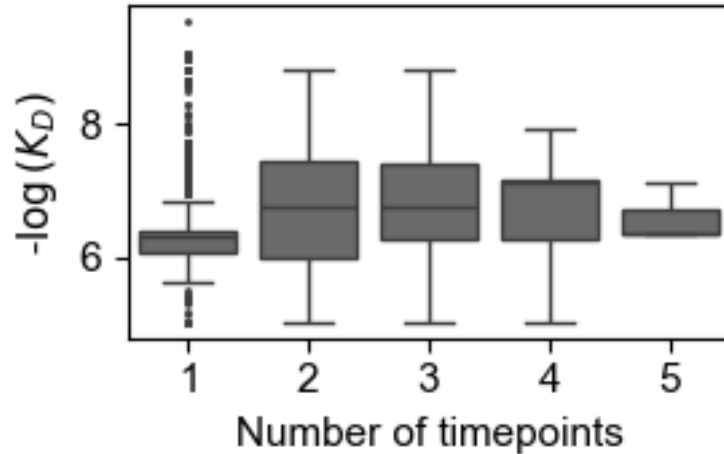
Binder lineages evolve differently

Evolutionary metrics differ
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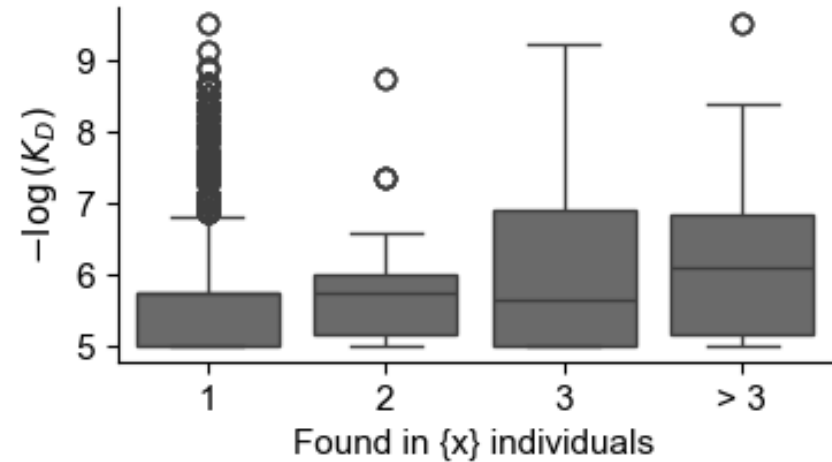
By contrast, no differences
between isotypes

The affinity of the germline sequence matters too !

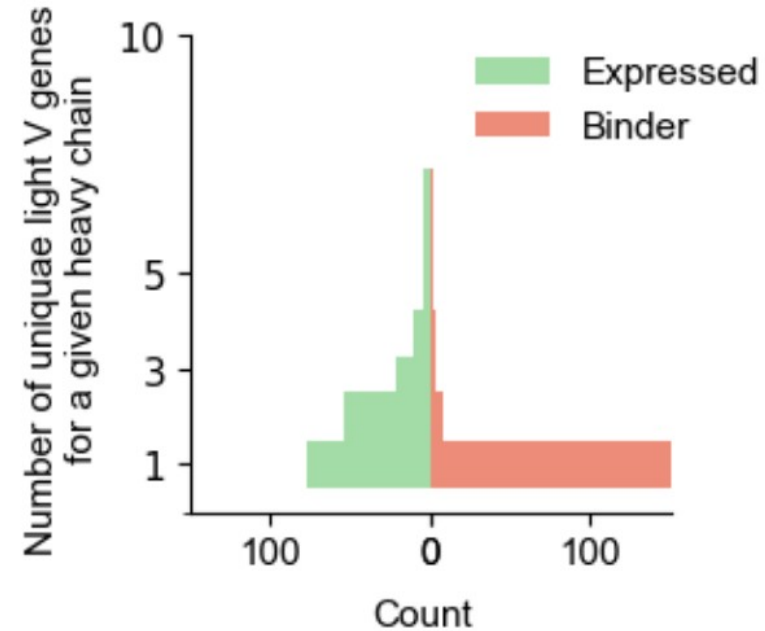
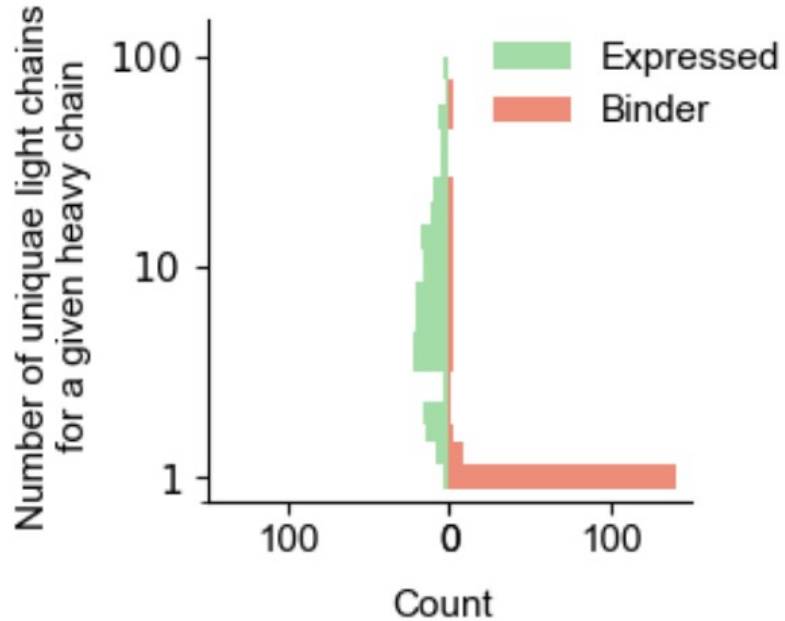


High affinity germline
are found across more
timepoints

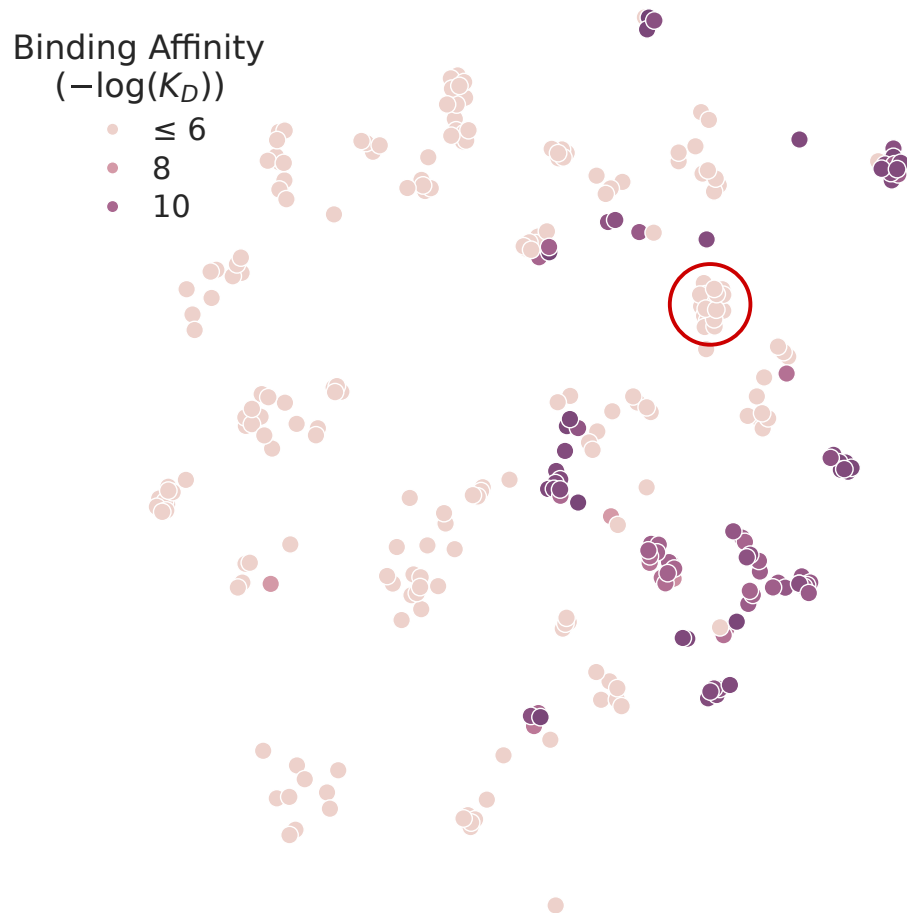
High affinity germline
are more shared



The light chain is crucial for specificity



Structural properties



AlphaFold is not doing very well here but is showing promise.

UMAP based on AlphaFold internal representation. Circled sequences are potential predicted binders.

- 10^9 B-cell sequences, 10^4 antibody-antigen interaction measured

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- We have characterized how the lineages that react to the antigen differ from the non-reacting lineages

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- We have characterized how the lineages that react to the antigen differ from the non-reacting lineages
- Repertoire are interesting, sequence them !
- In the future : more antigens, more targeted datasets (single-cell), other parameters of the interaction (k_{off} , resistance under forces)



We are recruiting up to two new team leaders.

Advancing Immunology Together

200 scientists • 16 research teams

State-of-the-art facilities:

Flow cytometry • Imaging • Genomics • Histology •
Computational biology • CRISPR screening

Start-up package:

Fully equipped lab • Salary • 2-year consumables funding •
Free core facilities

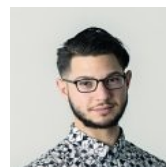
Application deadline: May 1st 2026

Contact: milpied@ciml.univ-mrs.fr

<https://ciml.univ-mrs.fr>

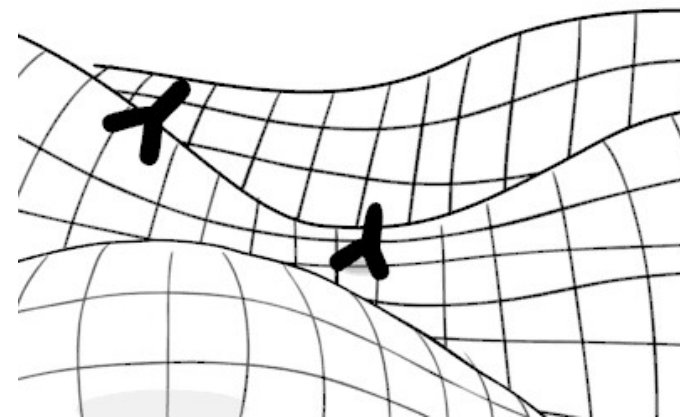
THANKS !

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Michael Desai
Alexandra Poret
Alief Moulana
Thierry Mora
Daniel Newton
Joao Ascensao
Berk Alpay
Aleksandra Walczak



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Charlotte Réocreux



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